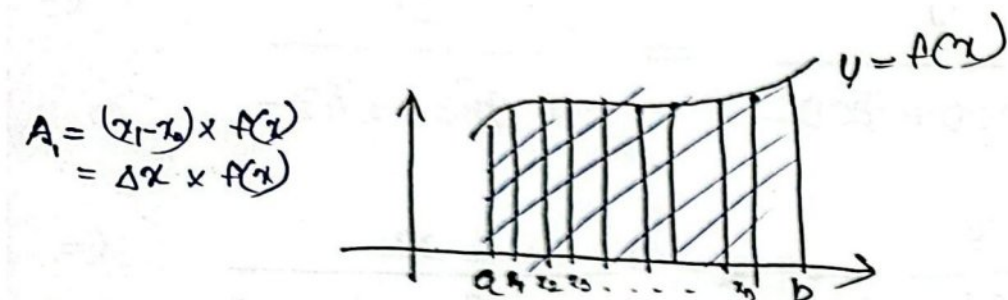


Lee-01
11/06/26

MAT-120
RHR

Topic: Riemann Sum:



Riemann Sum is an approximation of the area.

Let, $y = f(x)$ be a function defined on the closed interval $[a, b]$

It involves the following steps:

- ① Partition the Interval: Divide the interval $[a, b]$ in n -subintervals $[x_{k-1}, x_k]$ of width $\Delta x_k = x_k - x_{k-1}$ where $a = x_0 < x_1 < x_2 < \dots < x_n = b$
- ② Choose a number x_k^* in each subintervals $[x_{k-1}, x_k]$
- ③ The n numbers $x_1^*, x_2^*, \dots, x_n^*$ are called the sample points
- ④ Calculate the value $f(x)$ at $x = x_k^*$, then the

sum
$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

This sum is called Riemann Sum.

Since the sample points x_k^* can be chosen arbitrarily some common choices are given by:

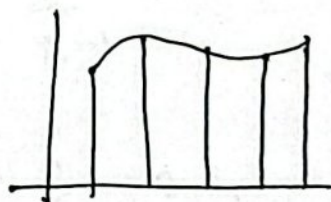
i) Left endpoint: $x_k^* = a + (k-1)\Delta x$

ii) Right endpoint: $x_k^* = a + k\Delta x$

iii) Midpoint: $x_k^* = a + (k - \frac{1}{2})\Delta x$

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

Q: Find the Riemann Sum of $f(x)$ on $[a, b]$ by using Left endpoint/Right/Mid (with 4 subintervals) (n of value 4)



Type-01

Q1 Find the area between the function of $f(x) = x-1$ at the interval $[0, 1]$ using

i) Left

ii) Right

iii) Mid point formula

$$1 + \dots + 8 + 11 = \sum_{k=1}^n$$

$$\frac{(1+11)11}{2} =$$

$$11 \dots 11 + 11 =$$

$$11$$

$$\sum_{k=1}^n$$

Soln: The length of each subintervals,
 $\Delta x = \frac{b-a}{n} = \frac{1}{n}$

For Left end point: $x_k^* = a + (k-1)\Delta x$

$$= 0 + (k-1)\frac{1}{n}$$

$$= \frac{k-1}{n}$$

$\therefore$ Then the Riemann Sum:

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = A$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n f\left(\frac{k-1}{n}\right) \frac{1}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\left(\frac{k-1}{n} \right)^2 - 1 \right] \frac{1}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{k^2}{n^2} - \frac{1}{n^2} - \frac{1}{n} \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} \sum_{k=1}^n$$

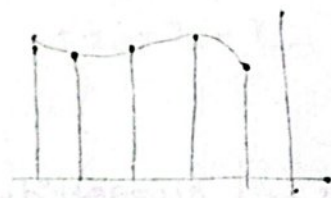
$$\Rightarrow \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \sum_{k=1}^n k^2 - \frac{1}{n^2} \sum_{k=1}^n 1 - \frac{1}{n} \sum_{k=1}^n 1 \right]$$

$$\therefore \sum_{k=1}^n k^2 \Rightarrow 1+2+3+\dots+n$$

$$= \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n 1 \Rightarrow 1+1+1+\dots+n$$

$$\Rightarrow n$$



$$\Rightarrow \lim_{n \rightarrow \infty} \left[\frac{1}{n^2} \frac{n(n+1)}{2} - \frac{1}{n^2} \cdot n - \frac{1}{n} \cdot n \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left[\frac{n+1}{2n} - \frac{1}{n} - 1 \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left[\frac{n}{2n} + \frac{1}{2n} - \frac{1}{n} - 1 \right]$$

$$\Rightarrow \left[\lim_{n \rightarrow \infty} \left[\frac{n}{2n} + \frac{1}{2n} - \frac{1}{n} - 1 \right] \right]$$

$$\Rightarrow \frac{1}{2} + 0 - 0 - 1$$

$$\Rightarrow -\frac{1}{2} \quad (\text{Ans:})$$

20-01-2022

2-09-2022

Find the limit

of the sequence

① Left and Right

② Right and Left

③ Mid

④ Left

⑤ Right

⑥ Mid

⑦ Left

⑧ Right

⑨ Mid

⑩ Left

⑪ Right

⑫ Mid

⑬ Left

⑭ Right

⑮ Mid

⑯ Left

⑰ Right

⑱ Mid

⑲ Left

⑳ Right

$$\sum_{k=1}^n k^2 \Rightarrow 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\Rightarrow \frac{n(n+1)(2n+1)}{6}$$

$$\Rightarrow \frac{n(n+1)(2n+1)}{6}$$

$$\Rightarrow \frac{n(n+1)(2n+1)}{6}$$

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$$\Rightarrow \frac{n(n+1)(2n+1)}{6}$$

Lec-02
 13/06/26

MAT120-RHR

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

Type-2
 n given

Q1 Find the Riemann Sum of $f(x) = 3x+1$ over $[2, 6]$ with 4 subintervals by using

- ① Left end point
- ② Right end point
- ③ Mid " "

Solⁿ: Left $\Delta x = \frac{b-a}{n} = \frac{6-2}{4} = 1$

For left end point: $x_k^* = a + (k-1)\Delta x$

if $k=1$ $x_1^* = 2 + (1-1) \cdot 1 = 2$

$k=2$ $x_2^* = 2 + (2-1) \cdot 1 = 3$

$k=3$ $x_3^* = 2 + (3-1) \cdot 1 = 4$

$k=4$ $x_4^* = 2 + (4-1) \cdot 1 = 5$

$f(x_1^*) = f(2) = 7$

$f(x_2^*) = f(3) = 10$

$f(x_3^*) = f(4) = 13$

$f(x_4^*) = f(5) = 16$

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x_k$$

$$= \sum_{k=1}^4 f(x_k^*) \Delta x$$

$$= f(x_1^*) \Delta x + f(x_2^*) \Delta x + f(x_3^*) \Delta x + f(x_4^*) \Delta x$$

$$= 7 + 10 \cdot 1 + 13 \cdot 1 + 16 \cdot 1$$

$$= 46 \text{ (Ans.)}$$

Antiderivative

Another method used to find the area below the curve $f(x)$ is using anti-derivative.

A function $F(x)$ is called an antiderivative of a function $f(x)$ on a given interval, if $F'(x) = f(x)$ $\forall$ values of x ,

It is written as,

$$\int f(x) dx = F(x) + C$$

where, C is an arbitrary constant.

Types of Integration:

$\Rightarrow$ Indefinite Integral: Used to find the general Antiderivative where the ~~begin~~ limit of integration are not defined. Which can be written as,

$$\int f(x) dx = F(x) + C$$

$\Rightarrow$ Definite Integral: Used to find area between two limits, where the limits of integration are given,

$$\int_a^b f(x) dx = F(b) - F(a)$$

where, $F(x)$ is an anti derivative of $f(x)$

Formulae:

$$\Rightarrow \int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\Rightarrow \int \frac{1}{x} dx = \ln|x| + C$$

$$\Rightarrow \int e^x dx = e^x + C$$

$$\Rightarrow \int e^{mx} dx = \frac{e^{mx}}{m} + C$$

$$\Rightarrow \int \sin x dx = -\cos x + C$$

$$\Rightarrow \int \cos x dx = \sin x + C$$

$$\Rightarrow \int dx = x + C$$

$$\Rightarrow \int \sec^2 x dx = \tan x + C$$

$$\Rightarrow \int \sec x \tan x dx = \sec x + C$$

$$\Rightarrow \int \csc x \cot x dx = -\csc x + C$$

$$\Rightarrow \int \csc x dx = -\ln|\csc x + \cot x| + C$$

$$\Rightarrow \int a^x dx = \frac{a^x}{\ln|a|} + C$$

* Continuity Implies Integrability:

$$\int_{-1}^3 \frac{1}{x^2} dx = \int_{-1}^3 x^{-2} dx = \int_{-1}^3 \frac{x^{-2+1}}{-2+1} dx$$

$$\Rightarrow \frac{x^{-2+1}}{-2+1} = -\frac{4}{3}$$

Methods:

$\Rightarrow$ Method of substitution: (u substitution)
If $u = g(x)$ is a differentiable function continuous on interval, then

$$\int f(g(x)) \cdot g'(x) dx = \int f(u) du = F(u) + C = F(g(x)) + C$$

* $u = g(x)$
 $\frac{du}{dx} = g'(x)$

Q1 $\int x^3 \cos(x^4+2) dx$ — (i) $\frac{1}{4} [\sin u] + C$

Soln: Let $u = x^4+2$
 $\frac{du}{dx} = 4x^3$
 $\Rightarrow \frac{du}{4} = x^3 dx$
 $\Rightarrow \int \cos(u) \cdot \frac{du}{4}$
 $\Rightarrow \frac{1}{4} \int \cos u du$

(i) $\Rightarrow \frac{1}{4} \sin(x^4+2) + C$

$\frac{1}{4} = \frac{ub}{rb}$
 $ub \delta = rb$

Q1 $\int \frac{x}{\sqrt{1-4x^2}} dx$ ——— (1)

Solⁿ: Let $u = 1 - 4x^2$

$$\frac{du}{dx} = -8x$$

$$\Rightarrow \frac{du}{-8} = x dx$$

$\therefore \int \frac{1}{\sqrt{u}} \left(-\frac{1}{8}\right) du$

$$\Rightarrow -\frac{1}{8} \int \frac{1}{\sqrt{u}} dx$$

$$\Rightarrow -\frac{1}{8} \left[\frac{u^{-1/2+1}}{-1/2+1} \right] + C$$

$$\Rightarrow -\frac{1}{8} \left[\frac{u^{1/2}}{1/2} \right] + C$$

$$\Rightarrow -\frac{1}{4} \sqrt{u} + C$$

$$\Rightarrow -\frac{1}{4} \sqrt{1-4x^2} + C$$

(Ans.)

Q1 $\int \frac{dx}{\left(\frac{x}{2} - 8\right)^5}$ ——— (1)

Let $u = \frac{x}{2} - 8$

$$\frac{du}{dx} = \frac{1}{2}$$

$$dx = 2 du$$

$$\int \frac{2 du}{u^5}$$

$$\Rightarrow 2 \int u^{-5} du$$

$$\Rightarrow 2 \left[\frac{u^{-5+1}}{-5+1} \right] + C$$

$$\Rightarrow 2 \left[\frac{u^{-4}}{-4} \right] + C$$

$$\Rightarrow -\frac{1}{2} u^{-4} + C$$

$$\Rightarrow -\frac{1}{2} \left(\frac{x}{2} - 8 \right)^{-4} + C$$

Q1 $\int \frac{e^{4x}}{x^2} dx$

Q1 $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

Q1 $\int x^3 \sqrt{x^4+5} dx$

Q1 $\int x^2 \sqrt{x-1} dx$

Q1 $\int \frac{\sin(\ln x)}{x} dx$

Q1 $\int (x^2+1)^2 \cdot 2x dx$

⊛ Integration By Parts :

The rule that corresponds to the product rule is called the rule for integration by parts,

$$\int uv dx = \cancel{\int u \frac{d}{dx} v} u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx$$

There is a technique that helps to choose u & v properly, is known as **LIATE** Method.

This acronym comes from first letter of the following types of function:

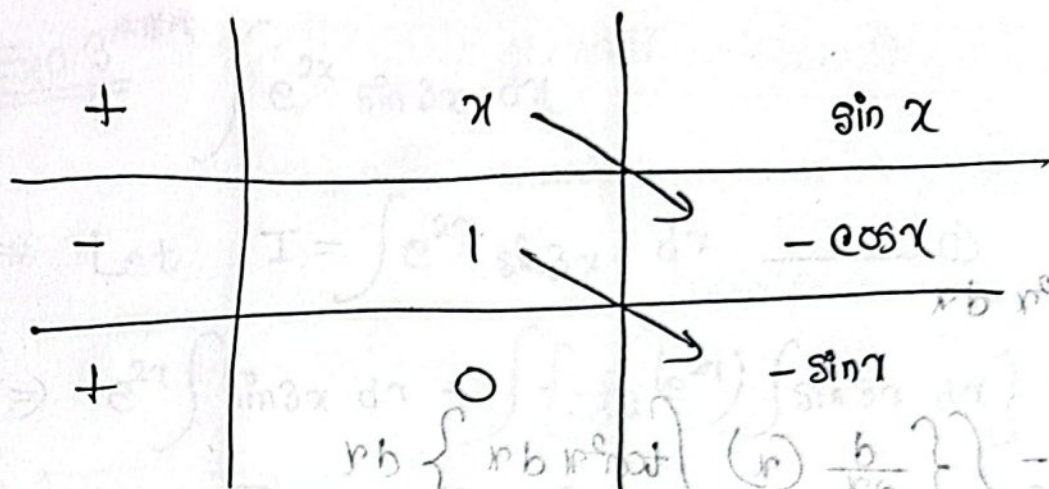
L $\Rightarrow$ Logarithmic

I $\Rightarrow$ Inverse

A $\Rightarrow$ Algebraic

T $\Rightarrow$ Trigonometric

E $\Rightarrow$ Exponential



$$\Rightarrow -x \cos x + \sin x \left\{ \frac{1}{x} \right\} - \sin x \left(1 - \frac{1}{x^2} \right)$$

Sheet:

Page: 11

Lab: tashfia.saad @bnu.ac.bd

$$(3.1) \text{ (A) } \frac{1}{x} \left[\frac{1}{x} - \cos x \right] - \left(1 - \frac{1}{x^2} \right) x$$

$$\left[\frac{1}{x^2} - \cos x \right] - \left(1 - \frac{1}{x^2} \right) x$$

$$\left\{ \frac{1}{x^2} - \cos x \right\} - \left(1 - \frac{1}{x^2} \right) x$$

$$\left\{ \left(\frac{1}{x^2} - \cos x \right) \right\} - \left[\frac{1}{x^2} - \cos x \right] x$$

$$\left[\frac{1}{x^2} - \cos x \right] \frac{1}{x} + \left[\frac{1}{x^2} - \cos x \right] \frac{1}{x}$$

$$\left[0 - 0 \right] \frac{1}{x} + \left[0 - x \right] \frac{1}{x}$$

$$(A) \frac{1}{x} - \cos x$$

lec-03
18/06/26

Exp 1 $\int x \tan^2 x \, dx$

$$\Rightarrow x \int \tan^2 x \, dx - \int \left\{ \frac{d}{dx} (x) \int \tan^2 x \, dx \right\} dx$$

$$\Rightarrow x \int (\sec^2 x - 1) \, dx - \int \{ 1 \cdot \int (\sec^2 x - 1) \, dx \} dx + x \cos x -$$

$$\Rightarrow x (\tan x - x) - \int (\tan x - x) \, dx$$

$$\Rightarrow x (\tan x - x) - \left[\ln(\sec x) - \frac{x^2}{2} \right] + C \quad (\text{Ans})$$

Formulae: $\int \tan x \, dx = \ln|\sec x|$

Exp 2 $\int_0^{\pi} x \sin 2x \, dx$

$$\Rightarrow x \int \sin 2x \, dx - \int \left\{ \frac{d}{dx} (x) \int \sin 2x \, dx \right\} dx$$

$$\Rightarrow x \left[-\frac{\cos 2x}{2} \right] - \int \left\{ 1 \cdot \left(-\frac{\cos 2x}{2} \right) \right\} dx$$

$$\Rightarrow -\frac{1}{2} x \cos 2x + \frac{1}{2} \int \cos 2x \, dx$$

$$\Rightarrow -\frac{1}{2} \left[x \cos 2x \right]_0^{\pi} + \frac{1}{2} \left[\frac{\sin 2x}{2} \right]_0^{\pi}$$

$$\Rightarrow -\frac{1}{2} [\pi - 0] + \frac{1}{4} [0 - 0]$$

$$\Rightarrow -\frac{\pi}{2} \quad (\text{Ans})$$

Exp 3*** $\int e^{2x} \sin 3x \, dx$

$\Rightarrow$ Let $I = \int e^{2x} \sin 3x \, dx$ ——— (i)

$\Rightarrow e^{2x} \int \sin 3x \, dx - \int \left\{ \frac{d}{dx} (e^{2x}) \int \sin 3x \, dx \right\} dx$

$\Rightarrow \boxed{e^{2x} \left[\frac{-\cos 3x}{3} \right] - \int \left\{ 2e^{2x} \left(-\frac{\cos 3x}{3} \right) \right\} dx}$

$\Rightarrow -\frac{1}{3} [e^{2x} \cos 3x] + \frac{2}{3} \int e^{2x} \cos 3x \, dx$ ——— (ii)

$\Rightarrow \int e^{2x} \cos 3x \, dx$

$\Rightarrow e^{2x} \int \cos 3x \, dx - \int \left\{ \frac{d}{dx} (e^{2x}) \int \cos 3x \, dx \right\} dx$

$\Rightarrow e^{2x} \left[\frac{\sin 3x}{3} \right] - \int \left\{ 2e^{2x} \left(\frac{\sin 3x}{3} \right) \right\} dx$

$\Rightarrow \frac{1}{3} [e^{2x} \sin 3x] - \frac{2}{3} \int e^{2x} \sin 3x \, dx$

$\Rightarrow \frac{1}{3} [e^{2x} \sin 3x] = \frac{2}{3} I$ [from (i)]

(ii) Hence,

$I = -\frac{1}{3} [e^{2x} \cos 3x] + \frac{2}{3} \left[\frac{1}{3} (e^{2x} \sin 3x) - \frac{2}{3} I \right]$

$\Rightarrow I = -\frac{1}{3} (e^{2x} \cos 3x) + \frac{2}{9} (e^{2x} \sin 3x) - \frac{4}{9} I$

$\Rightarrow I + \frac{4}{9} I = -\frac{1}{3} (e^{2x} \cos 3x) + \frac{2}{9} (e^{2x} \sin 3x)$

$\Rightarrow \frac{9I + 4I}{9} = -\frac{1}{3} (e^{2x} \cos 3x) + \frac{2}{9} (e^{2x} \sin 3x)$

$\Rightarrow I = \frac{9}{13} \left[-\frac{1}{3} e^{2x} \cos 3x + \frac{2}{9} e^{2x} \sin 3x \right] + C \text{ (Ans)}$

Reduction Formula

Exp (1)

Obtain the reduction formula for,

$$\int \sin^n x \, dx, \quad \text{Hence, evaluate}$$

$$\int \sin^6 x \, dx$$

Soln: Let $I_n = \int \sin^n x \, dx$

$$\Rightarrow \int \sin^{n-1} x \cdot \sin x \, dx$$

$$\Rightarrow \sin^{n-1} x \int \sin x \, dx - \int \left\{ \frac{d}{dx} (\sin^{n-1} x) \int \sin x \, dx \right\} dx$$

$$\Rightarrow \sin^{n-1} x \cdot (-\cos x) - \int \left\{ (n-1)(\sin x)^{n-2} \cdot \cos x (-\cos x) \right\} dx$$

$$\Rightarrow -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x \, dx$$

$$\Rightarrow -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) \, dx$$

$$\Rightarrow -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - \int \sin^n x \, dx$$

$$\Rightarrow -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx$$

$$\Rightarrow [-\cos x \sin^{n-1} x + (n-1) I_{n-2} - (n-1) I_n] = I$$

$$\Rightarrow I_n + (n-1) I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2}$$

$$\Rightarrow I_n (1 + n - 1) = -\cos x \sin^{n-1} x + (n-1) I_{n-2}$$

$$\Rightarrow n I_n = -\cos x \sin^{n-1} x + (n-1) I_{n-2}$$

$$\Rightarrow I_n = \frac{-\cos x \sin^{n-1} x}{n} + \frac{(n-1)}{n} I_{n-2}$$

Now,

$$I_n = -\frac{\cos x \sin^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$$

$n=6$ ବାଛି,

$$I_6 = -\frac{\cos x \sin^5 x}{6} + \frac{5}{6} I_4$$

$$I_4 = -\frac{\cos x \sin^3 x}{4} + \frac{3}{4} I_2$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x + \frac{15}{24} I_2$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x + \frac{15}{24} \left[-\frac{\cos x \sin x}{2} + \frac{1}{2} I_0 \right]$$

$$\int \sec^n x dx, \int \sec^5 x dx$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x - \frac{15}{48} \cos x \sin x + \frac{15}{48} x + C$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x - \frac{5}{16} \cos x \sin x + \frac{5}{16} x + C$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x - \frac{5}{16} \cos x \sin x + \frac{5}{16} x + C$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x - \frac{5}{16} \cos x \sin x + \frac{5}{16} x + C$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x - \frac{5}{16} \cos x \sin x + \frac{5}{16} x + C$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x - \frac{5}{16} \cos x \sin x + \frac{5}{16} x + C$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x - \frac{5}{16} \cos x \sin x + \frac{5}{16} x + C$$

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$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x - \frac{5}{16} \cos x \sin x + \frac{5}{16} x + C$$

$$= -\frac{1}{6} \cos x \sin^5 x - \frac{5}{24} \cos x \sin^3 x - \frac{5}{16} \cos x \sin x + \frac{5}{16} x + C$$

Lec-04

Type-01

Trigonometric Integrals:

$\int \sin^m x \cos^n x dx$	Procedure	Identities
① n odd	① split of a factor of $\cos x$ ② substitution, $z = \sin x$	$\cos^2 x = 1 - \sin^2 x$
② m odd	① " " " " " $\sin x$ ② " " " " $z = \cos x$	$\sin^2 x = 1 - \cos^2 x$
③ m even ④ n even	Use the relevant identities to reduce the power of $\sin x$ & $\cos x$.	$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$ $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$

$$\begin{aligned}
 \text{Q1 } & \int \sin^4 x \cos^5 x dx \\
 & \Rightarrow \int \sin^4 x \cos^4 x \cos x dx \\
 & \Rightarrow \int \sin^4 x (1 - \sin^2 x)^2 \cos x dx \\
 & \Rightarrow \text{Let } z = \sin x,
 \end{aligned}$$

$$\frac{dz}{dx} = \cos x$$

$$dz = \cos x dx$$

$$\Rightarrow \int \sin^4 x (1 - \sin^2 x)^2 dz$$

$$\Rightarrow \int z^4 (1 - z^2 + z^4) dz$$

$$\Rightarrow \int (z^4 - 2z^6 + z^8) dz$$

$$\Rightarrow \int \frac{z^5}{5} - 2\frac{z^7}{7} + \frac{z^9}{9}$$

$$\Rightarrow \frac{\sin^5 x}{5} - \frac{2}{7} (\sin^7 x) + \frac{1}{9} (\sin^9 x)$$

Exp-02

$$\int \sin^5 x \cos^4 x \, dx$$

$$\Rightarrow \int \sin^4 x \sin x \cos^4 x \, dx$$

$$\Rightarrow \int (1 - \cos^2 x)^2 \sin x \cos^4 x \, dx$$

Let,

$$\cos x = z$$

$$\frac{dz}{dx} = -\sin x$$

$$dz = -\sin x \, dx$$

$$\Rightarrow - \int (1 - z^2)^2 \cdot z^4 \, dz$$

$$\Rightarrow - \int (1 - 2z^2 + z^4) z^4 \, dz$$

$$\Rightarrow - \int z^4 - 2z^6 + z^8 \, dz$$

$$\Rightarrow - \left(\frac{z^5}{5} - 2 \cdot \frac{z^7}{7} + \frac{z^9}{9} \right)$$

$$\Rightarrow - \frac{z^5}{5} + \frac{2}{7} z^7 - \frac{z^9}{9}$$

$$\Rightarrow - \frac{\cos^5 x}{5} + \frac{2}{7} \cos^7 x - \frac{\cos^9 x}{9}$$

$$\Rightarrow \left[\frac{2}{7} \cos^7 x - \frac{\cos^9 x}{9} - \frac{\cos^5 x}{5} \right] + C$$

$$\Rightarrow \frac{2}{7} \cos^7 x - \frac{\cos^9 x}{9} - \frac{\cos^5 x}{5} + C$$

Exp $\int \sin 4x \, dx$

$$\Rightarrow \int (\sin^2 x)^2 \, dx$$

$$\Rightarrow \int \left(\frac{1 - \cos 2x}{2} \right)^2 \, dx$$

$$\Rightarrow \frac{1}{4} \int (1 - \cos 2x) \, dx$$

$$\Rightarrow \frac{1}{4} \int [1 - 2\cos 2x + \cos^2 2x] \, dx$$

$$\Rightarrow \frac{1}{4} \int \left[1 - 2\cos 2x + \frac{1}{2} (1 + \cos 2 \cdot 2x) \right] \, dx$$

$$\Rightarrow \frac{1}{4} \int \left[1 - 2\cos 2x + \frac{1}{2} + \frac{1}{2} \cos 4x \right] \, dx$$

$$\Rightarrow \frac{1}{4} \left[x - 2 \frac{\sin 2x}{2} + \frac{1}{2} x + \frac{1}{2} \frac{\sin 4x}{4} \right] + C$$

Type-2

$$\int \sin(m\theta) \cos(n\theta) \, d\theta$$

Formulae:

i) $\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)]$

ii) $\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$

iii) $\cos A \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$

iv) $\cos A \sin B =$

Exp

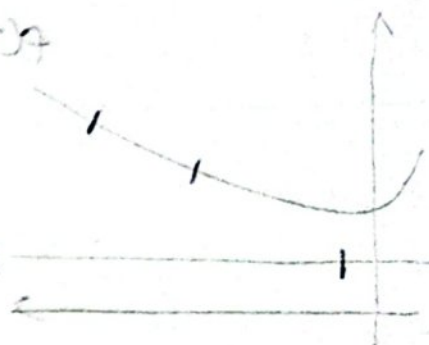
$$\int \sin 4x \cos 5x \, dx \quad \frac{-ICRITAM}{10-001}$$

282014

$$\Rightarrow \int \frac{1}{2} [\sin(9x) + \sin(-x)] \, dx$$

$$\Rightarrow \int \frac{1}{2} [\sin 9x - \sin x] \, dx$$

$$\Rightarrow \frac{1}{2} \left[-\frac{\cos 9x}{9} - (-\cos x) \right] + C$$



Exp

$$\int \sin 5x \sin 7x \, dx, \int \cos 7x \cos 9x$$

ICRITAM

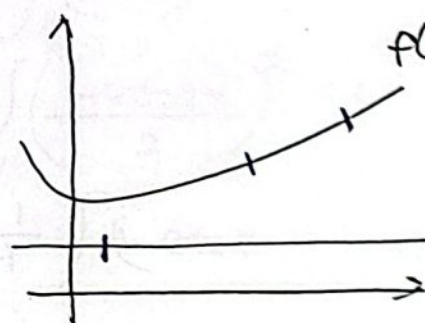
282014

Integration of the form $\int \sin^m x \cos^n x \, dx$ where m is odd and n is even.
 Case (i): n is even $\rightarrow$ Apply $\sin^2 x = 1 - \cos^2 x$ and use substitution $u = \cos x$.
 Case (ii): m is odd $\rightarrow$ Apply $\cos^2 x = 1 - \sin^2 x$ and use substitution $u = \sin x$.
 Case (iii): Both m and n are even $\rightarrow$ Use double angle formulas and reduction formulas.

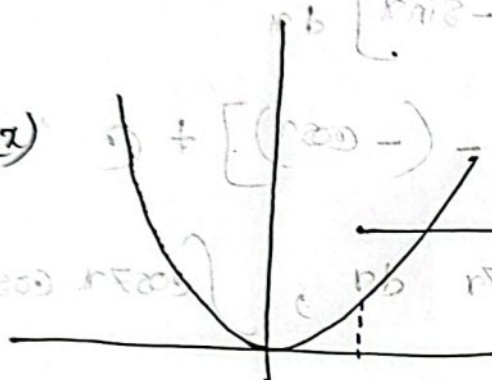
4U288

MAT120L

Lec-01



$f(x)$ Piece-wise function



Lec-05
25/06/26

MAT120

* Integrating Products of Tangents & secants
 $\Rightarrow$ Integration of the form $\int \tan^m x \cdot \sec^n x \, dx$

Case (i): n even $\rightarrow$

- ① split of, $\sec^2 x$
- ② Apply, $\sec^2 x = 1 + \tan^2 x$
- ③ $z = \tan x$

Case (ii): m odd $\rightarrow$

- ① split of $\sec x \tan x$
- ② Apply, $\tan^2 x = \sec^2 x - 1$
- ③ $z = \sec x$

less imp
case (iii)

m, n even $\rightarrow$ ① use $\tan^2 x = \sec^2 x - 1$ to reduce the power & finally use the reduction formula

Exp-01

$$\int \tan^2 x \sec^3 x dx$$

Soln:

$$\int \tan^2 x \tan x \sec^3 x dx$$

$$\Rightarrow \int \tan^2 x (\sec^2 x - 1) \sec^2 x \sec x \tan x dx \quad \text{--- ①}$$

Let, $z = \sec x$
 $dz = \sec x \tan x dx$

① $\rightarrow$ 266,

$$\int (z^2 - 1) z^2 dz$$

$$\Rightarrow \int (z^4 - z^2) dz$$

$$\Rightarrow \frac{z^5}{5} - \frac{z^3}{3} \Rightarrow \frac{\sec^5 x}{5} - \frac{\sec^3 x}{3} + C$$

Exp-2

$$\int \tan^3 x \sec^2 x dx$$

$$\Rightarrow \int \tan^3 x (\tan^2 x + 1) dx$$

$$\Rightarrow \int \tan^3 x z^2 dz$$

$$\Rightarrow \frac{z^4}{4} \Rightarrow \frac{\tan^4 x}{4} + C$$

Let $z = \tan x$

$$dz = \sec^2 x dx$$

T-04 Trigonometric Substitution

Integration of the form

	Substitutions	Simplification
$\int \sqrt{a^2 - x^2} dx$	$x = a \sin \theta / a \cos \theta$	$a \sin \theta / a \cos \theta$
$\int \sqrt{a^2 + x^2} dx$	$x = a \tan \theta$	$a \sec \theta$
$\int \sqrt{x^2 - a^2} dx$	$x = a \sec \theta$	$a \tan \theta$

Exp 1

$$\int \frac{\sqrt{9-x^2}}{x^2} dx \quad \text{--- (1)}$$

$$\Rightarrow \int \frac{1}{x^2} \sqrt{3^2 - x^2} dx$$

$$\Rightarrow \int \frac{1}{(3 \sin \theta)^2} \sqrt{9 - (3 \sin \theta)^2}$$

$$\Rightarrow \int \frac{\sqrt{9(1 - \sin^2 \theta)}}{9 \sin^2 \theta} 3 \cos \theta d\theta$$

$$\Rightarrow \int \frac{\sqrt{\cos^2 \theta}}{\sin^2 \theta} \cos \theta d\theta$$

$$\Rightarrow \int \frac{\cos^2 \theta}{\sin^2 \theta} d\theta$$

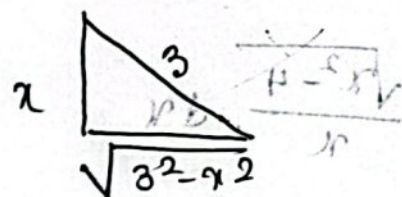
$$\Rightarrow \int \cot^2 \theta d\theta \Rightarrow \int (\csc^2 \theta - 1) d\theta \Rightarrow -\cot \theta - \theta$$

$$\text{Let } x = 3 \sin \theta$$

$$dx = 3 \cos \theta d\theta$$

$$\Rightarrow \frac{-\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + C$$

$$\sin \theta = \frac{x}{3}$$



$$\cot \theta = \frac{\sqrt{3^2-x^2}}{x}$$

$$\text{Q1} \int \frac{1}{x^2 \sqrt{x^2+4}} dx$$

Let,

$$\Rightarrow \int \frac{1}{(2 \tan \theta)^2 \sqrt{4 + (2 \tan \theta)^2}}$$

$$x = 2 \tan \theta$$

$$dx = 2 \sec^2 \theta d\theta$$

$$= \int \frac{1}{4 \tan^2 \theta \sqrt{4 + 4 \tan^2 \theta}} \cdot 2 \sec^2 \theta d\theta$$

$$= \int \frac{1}{4 \tan^2 \theta \sqrt{4(1 + \tan^2 \theta)}} \cdot 2 \sec^2 \theta d\theta$$

$$= \int \frac{1}{4 \tan^2 \theta \sqrt{4 \sec^2 \theta}} \cdot 2 \sec^2 \theta d\theta$$

$$= \int \frac{2 \sec^2 \theta}{4 \tan^2 \theta \sqrt{4 \sec^2 \theta}} d\theta$$

$$= \frac{1}{4} \int \frac{\sec^2 \theta d\theta}{\tan^2 \theta \sqrt{\sec^2 \theta}}$$

$$= \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta$$

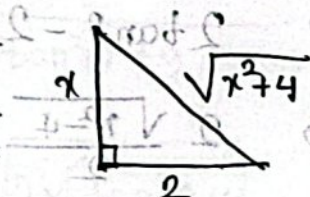
$$= \frac{1}{4} (-\operatorname{cosec} \theta) + C$$

Now,

$$\tan \theta = \frac{x}{2}$$

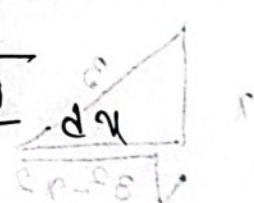
$$\therefore \operatorname{cosec} \theta = \frac{\sqrt{x^2+4}}{x}$$

$$\therefore \frac{1}{4} \left(-\frac{\sqrt{x^2+4}}{x} \right) + C$$



Exp-3

$$\frac{r}{2} = \sec \theta$$

$$\int \frac{\sqrt{x^2 - 4}}{x} dx$$


$$\sec \theta = \frac{x}{2} \Rightarrow \frac{x}{2} = \sec \theta$$

Let,

$$x = 2 \sec \theta$$

$$dx = 2 \tan \theta \sec \theta d\theta$$

$$\Rightarrow \int \frac{\sqrt{(2 \sec \theta)^2 - 4}}{2 \sec \theta} \cdot 2 \tan \theta \sec \theta d\theta$$

$$\Rightarrow \int \frac{\sqrt{4 \sec^2 \theta - 4}}{2 \sec \theta} \cdot 2 \tan \theta \sec \theta d\theta$$

$$\Rightarrow \int \frac{\sqrt{4(\sec^2 \theta - 1)}}{2 \sec \theta} \cdot 2 \tan \theta \sec \theta d\theta$$

$$\Rightarrow \int 2 \sqrt{\sec^2 \theta - 1} \tan \theta d\theta$$

$$\Rightarrow 2 \int 2 \tan^2 \theta d\theta$$

$$\Rightarrow 2 \int \tan^2 \theta d\theta$$

$$\Rightarrow 2 \int (\sec^2 \theta - 1) d\theta$$

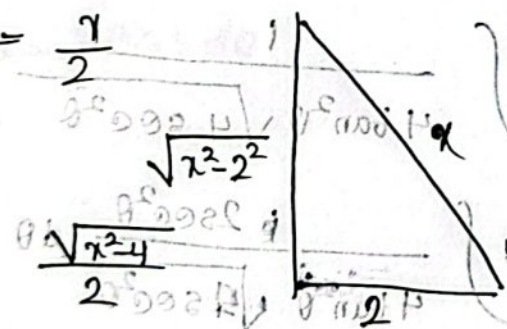
$$\Rightarrow 2 \int (\tan \theta - \theta)$$

$$\Rightarrow 2 \tan \theta - 2\theta$$

$$\Rightarrow \frac{2 \sqrt{x^2 - 4}}{2} - 2 \left(\sec^{-1} \left(\frac{x}{2} \right) \right)$$

$$\sec \theta = \frac{x}{2}$$

$$\tan \theta = \frac{\sqrt{x^2 - 4}}{2}$$



$$\frac{1}{\sec \theta} = \frac{2}{x}$$

$$\frac{1}{\sec \theta} = \frac{2}{x}$$

$$\frac{1}{\sec \theta} = \frac{2}{x}$$

QUIZ-01

02/07/26

⇒ Integration by Parts

⇒ Trigonometric Integral

⇒ T1 Substitution

⇒ Riemann Sum

* Trigonometric Substitutions:

T-5

Square root of Quadratic

$$\text{Exp-01} \quad \frac{0}{(x-1)} + \frac{B}{(x-1)} + \frac{A}{(x-1)} = \frac{1}{(x-1)}$$

$$\begin{aligned} & \int \frac{\sqrt{4x^2 - 16x + 52}}{(x-1)} dx \\ &= \int \sqrt{4(x^2 - 4x + 13)} dx \\ &= 2 \int \sqrt{x^2 - 4x + 13} dx \\ &= 2 \int \sqrt{x^2 - 2 \cdot x \cdot 2 + 2^2 + 9} dx \\ &= 2 \int \sqrt{(x-2)^2 + 3^2} dx \\ &= 2 \int \sqrt{(x-2)^2 + 3^2} dx \end{aligned}$$

Let, $z = x - 2$

$$\frac{dz}{dx} = 1$$

Ex-01 $dz = dx$

$$2 \int \sqrt{z^2 + 3^2} dz$$

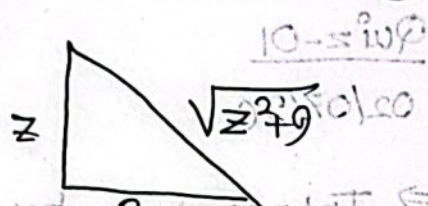
$$\Rightarrow 2 \int \sqrt{(3 \tan \theta)^2 + 3^2}$$

$$\Rightarrow 2 \int \sqrt{9 \tan^2 \theta + 9}$$

$$\Rightarrow 2 \int 3 \sqrt{\tan^2 \theta + 1} \\ \Rightarrow 6 \int \sec \theta d\theta$$

$$\Rightarrow 18 \left[\frac{1}{2} \sec \theta \tan \theta + \frac{1}{2} \ln |\sec \theta + \tan \theta| \right] \quad (\text{Reduction Formula}) \quad \text{--- (11)}$$

$$\Rightarrow 18 \left[\frac{1}{2} \cdot \frac{\sqrt{z^2+9}}{3} \cdot \frac{z}{3} + \frac{1}{2} \ln \left| \frac{\sqrt{z^2+9}}{3} + \frac{z}{3} \right| \right]$$



$$\Rightarrow 18 \left[\frac{1}{18} \left(\sqrt{(x-2)^2+9} (x-2) + \frac{1}{2} \ln \left| \frac{\sqrt{(x-2)^2+9}}{3} + \frac{x-2}{3} \right| \right) \right]$$

* Integrating Rational Functions by Partial Fraction :-

i) Distinct Root: $\frac{1}{(x-a)(x-b)} \equiv \frac{A}{x-a} + \frac{B}{x-b}$

ii) Repeated Root: $\frac{1}{(x-a)^2(x-b)} \equiv \frac{A}{x-a} + \frac{B}{(x-a)^2} + \frac{C}{x-b}$

iii) Quadratic Root: $\frac{1}{(ax^2+bx+c)(x-a)(x-b)} \equiv \frac{Ax+B}{ax^2+bx+c} + \frac{C}{x-a} + \frac{D}{x-b}$

Ex-01 Evaluate $\int \frac{5x-10}{(x-4)(x+1)} dx$

$\Rightarrow$ Now, $\frac{5x-10}{(x-4)(x+1)} \equiv \frac{A}{x-4} + \frac{B}{x+1}$

$\Rightarrow \frac{5x-10}{(x-4)(x+1)} \equiv \frac{A(x-4)(x+1)}{(x-4)(x+1)} + \frac{B(x-4)(x+1)}{(x-4)(x+1)}$

$\Rightarrow \frac{5x-10}{(x-4)(x+1)} \equiv \frac{A(x+1)}{(x-4)(x+1)} + \frac{B(x-4)}{(x-4)(x+1)}$

$\Rightarrow \frac{5x-10}{(x-4)(x+1)} \equiv \frac{Ax+A+Bx-4B}{(x-4)(x+1)}$

$$\Rightarrow 5x-10 \equiv x(A+B) + A-4B \equiv 1-10x+x$$

Equating coefficients of x & constant term

$$\left. \begin{array}{l} x: \quad 5 = A+B \quad \text{--- (i)} \\ \text{Cons} \quad -10 = A-4B \quad \text{--- (ii)} \end{array} \right\} \begin{array}{l} A=2 \\ B=3 \end{array}$$

$$\Rightarrow \int \left[\frac{2}{(x-4)} + \frac{3}{(x+1)} \right] dx$$

$$\Rightarrow \frac{2 \ln(x-4)}{(x-4)} + \frac{3 \ln(x+1)}{(x+1)} + \frac{0}{(x-4)} = \frac{1-10x+x}{(x-4)(x+1)}$$

Exp-02

$$\int \frac{x^2+2x-1}{2x^3+3x^2-2x} \left[\frac{1}{dx} \frac{1}{x} - \frac{1}{(x-2)} \frac{1}{x} + \frac{1}{x} \frac{1}{x} \right]$$

$$\Rightarrow \int \frac{x^2+2x-1}{x(2x^2+3x-2)} dx = \frac{1}{x} \frac{1-10x+x}{(x-4)(x+1)} + \frac{1}{x} \frac{1}{x}$$

$$\Rightarrow \int \frac{x^2+2x-1}{x(2x-1)(x+2)} dx$$

$$\Rightarrow \int \frac{x^2+2x-1}{x(2x-1)(x+2)} \equiv \frac{A}{x} + \frac{B}{(2x-1)} + \frac{C}{x+2}$$

$$\begin{aligned} \Rightarrow x^2+2x-1 &\equiv A(2x-1)(x+2) + B(x+2)(x) + C(2x-1)(x) \\ &\equiv A(2x^2-x+4x-2) + B(x^2+2x) + C(2x^2-x) \\ &\equiv A(2x^2+3x-2) + B(x^2+2x) + C(2x^2-x) \end{aligned}$$

$$x^2 + 2x - 1 \equiv A(2x^2 + 3x - 2) + B(x^2 + 2x) + C(2x^2 - x)$$

$$\Rightarrow 2x^2A + 3xA - 2A + Bx^2 + 2xB + 2x^2C - Cx$$

$$\Rightarrow x^2(2A + B + 2C) + x(3A + 2B - C) - 2A$$

$$\Rightarrow \begin{cases} 1 \equiv 2A + B + 2C \\ 2 \equiv 3A + 2B - C \\ -1 \equiv -2A \end{cases} \quad \begin{cases} A = \frac{1}{2} \\ B = \frac{9}{5} \\ C = -\frac{1}{10} \end{cases}$$

$$\Rightarrow \frac{x^2 + 2x - 1}{x(2x-1)(x+2)} \equiv \frac{1}{2(2)} + \frac{9}{5(2x-1)} - \frac{1}{10(x+2)}$$

$$\Rightarrow \int \left[\frac{1}{2} \frac{1}{x} + \frac{1}{5} \frac{1}{(2x-1)} - \frac{1}{10} \frac{1}{(x+2)} \right] \frac{1-xe^{2x}}{xe^{2x}-e^{2x}+e^{2x}}$$

$$\Rightarrow \frac{1}{2} \ln|x| + \frac{1}{5} \ln \frac{2x-1}{2} - \frac{1}{10} \ln|x+2| + \frac{e^x}{(e-xe^{2x})x}$$

Ex-03

$$\int \frac{2x^2 - x + 4}{x^3 + 4x} dx$$

$$\Rightarrow \int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx = \frac{1-xe^{2x}}{(e+x)(1-xe)x}$$

$$\Rightarrow \frac{2x^2 - x + 4}{x(x^2 + 4)} \equiv \frac{Ax + B}{x^2 + 4} + \frac{C}{x} \equiv \frac{1-xe^{2x}}{(e+x)(1-xe)x}$$

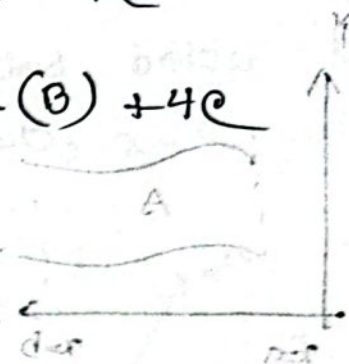
$$(x^2 - 4C) + (Ax + B) + (e - xe^{2x})A \equiv$$

$$\Rightarrow 2x^2 - x + 4 \equiv (Ax+B)x + C(x^2+4)$$

$$\Rightarrow 2x^2 - x + 4 \equiv Ax^2 + Bx + Cx^2 + 4C$$

$$\Rightarrow x^2(A+C) + x(B) + 4C$$

$$\Rightarrow \left. \begin{aligned} 2 &\equiv A+C \\ -1 &\equiv B \\ 4 &\equiv 4C \end{aligned} \right\} \begin{aligned} A &= 1 \\ B &= -1 \\ C &= 1 \end{aligned}$$



$$\Rightarrow \int \frac{1x-1}{x^2+4} + \frac{1}{x}$$

$$\Rightarrow \int \frac{x}{x^2+4} - \frac{1}{x^2+4} + \frac{1}{x}$$

$$\Rightarrow \ln|x| + \frac{1}{2} \ln|x^2+4| - \frac{1}{2} \tan^{-1} \frac{x}{2} + C$$

$$\frac{2}{8} = \frac{5}{8} = \frac{6}{9}$$

$$\Rightarrow \frac{2x}{8x^2} = \frac{5x^2}{8x^3} = \frac{6x^3}{9x^4+3x^2+2x+1}$$

*

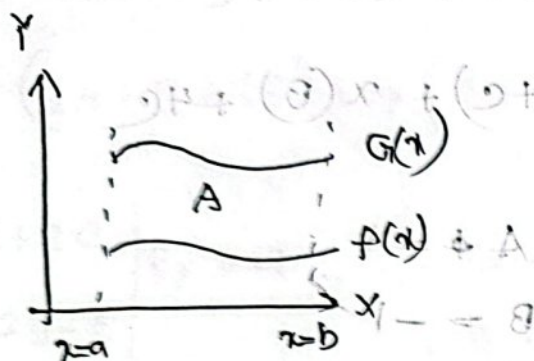
$$(y) < (y) \text{ \& } [y]$$

$$[y] - [y] = A$$

(Lec-07)

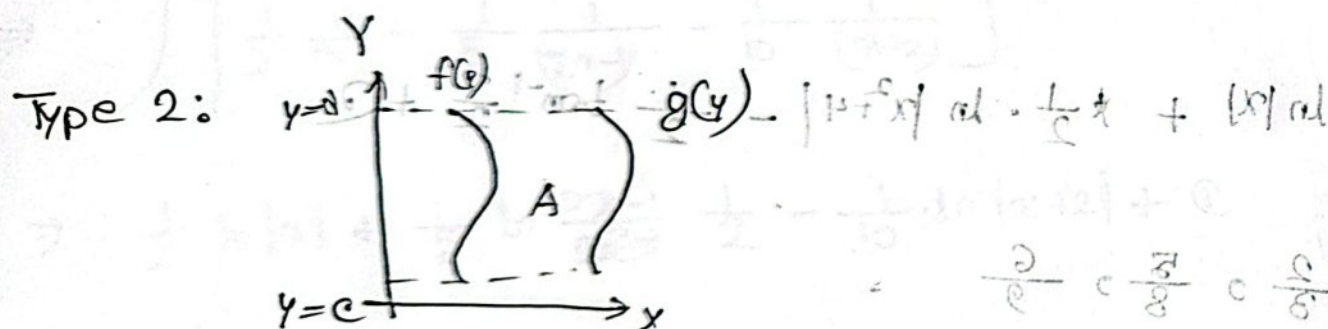
Area Between two curves:

Type-01



Suppose that $f(x)$ & $g(x)$ are continuous on $[a, b]$ &
 $g(x) \geq f(x)$

Then $A = \int_a^b [g(x) - f(x)] dx$



Type 2:

Suppose that $f(y)$ & $g(y)$ are continuous on
 $[c, d]$ & $g(y) \geq f(y)$

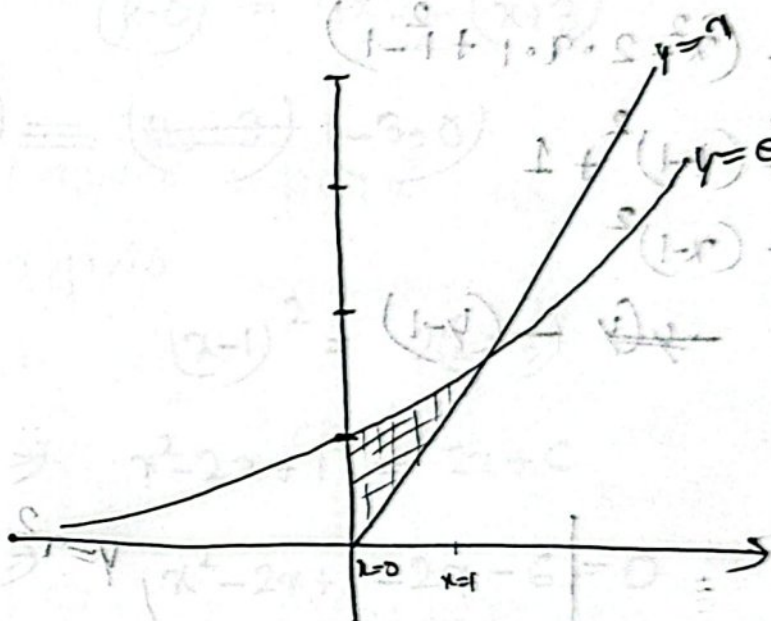
Then,

$$A = \int_c^d [g(y) - f(y)] dy$$

Exp-01

Sketch & find the area of the region bounded above by $y = e^x$ & bounded below by $y = x$ & bounded by the sides ($x=0, x=1$)

Soln

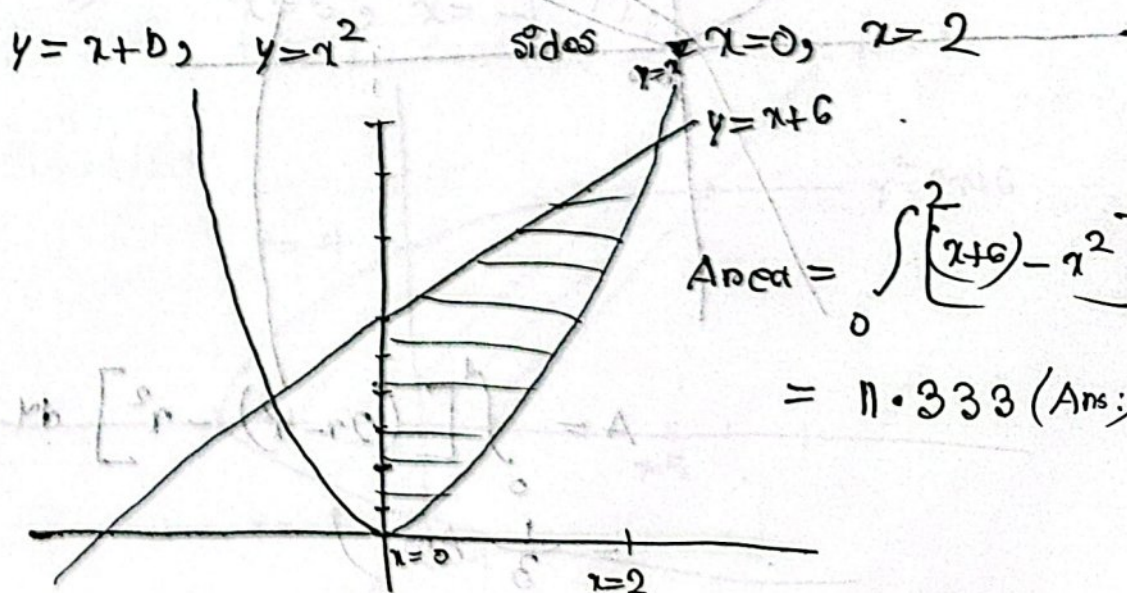


$$A = \int_{x=0}^1 [e^x - x] dx$$

$$= 1.21828$$

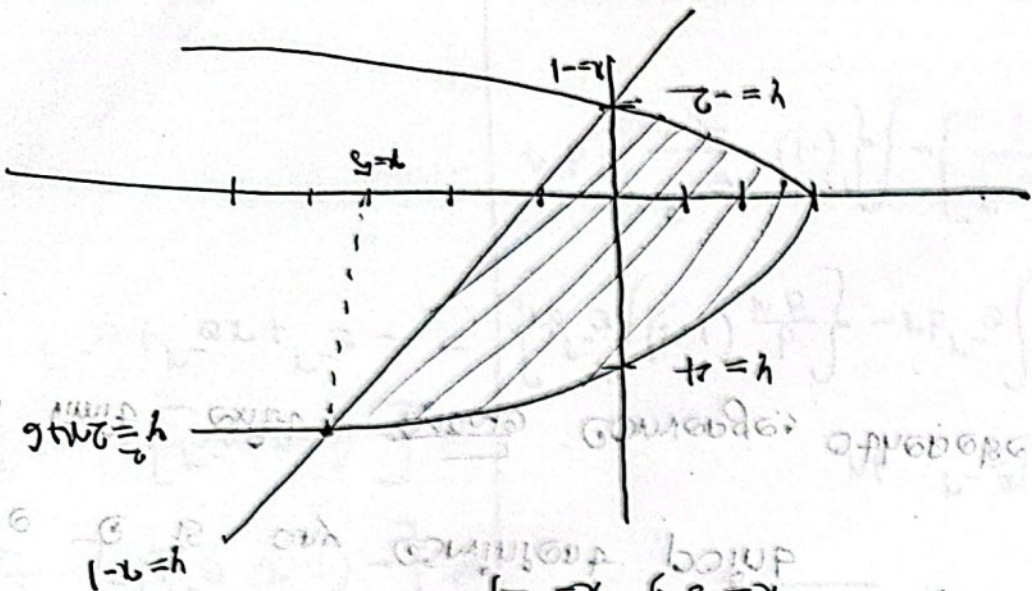
Exp-02

Soln:



$$\text{Area} = \int_0^2 [(x+6) - x^2] dx$$

$$= 11.333 \text{ (Ans.)}$$



$$\Rightarrow \int_0^2 (x^2 - 2x + 1 - (x - 1)) dx = 0$$

$$\Rightarrow \int_0^2 (x^2 - 3x + 2) dx = 0$$

$$\Rightarrow \int_0^2 (x^2 - 3x + 2) dx = 0$$

$$\Rightarrow \int_0^2 (x^2 - 3x + 2) dx = 0$$

$$(x-1)^2 = 2(x+3)$$

$$(x-1)^2 = 2(x+3)$$

$$(x-1)^2 = 2(x+3)$$

$$(x-1)^2 = 2(x+3)$$

$$(x-1)^2 = 2(x+3)$$

$$y = x-1, y^2 = 2x+6$$

$$E_{xp}(04)$$

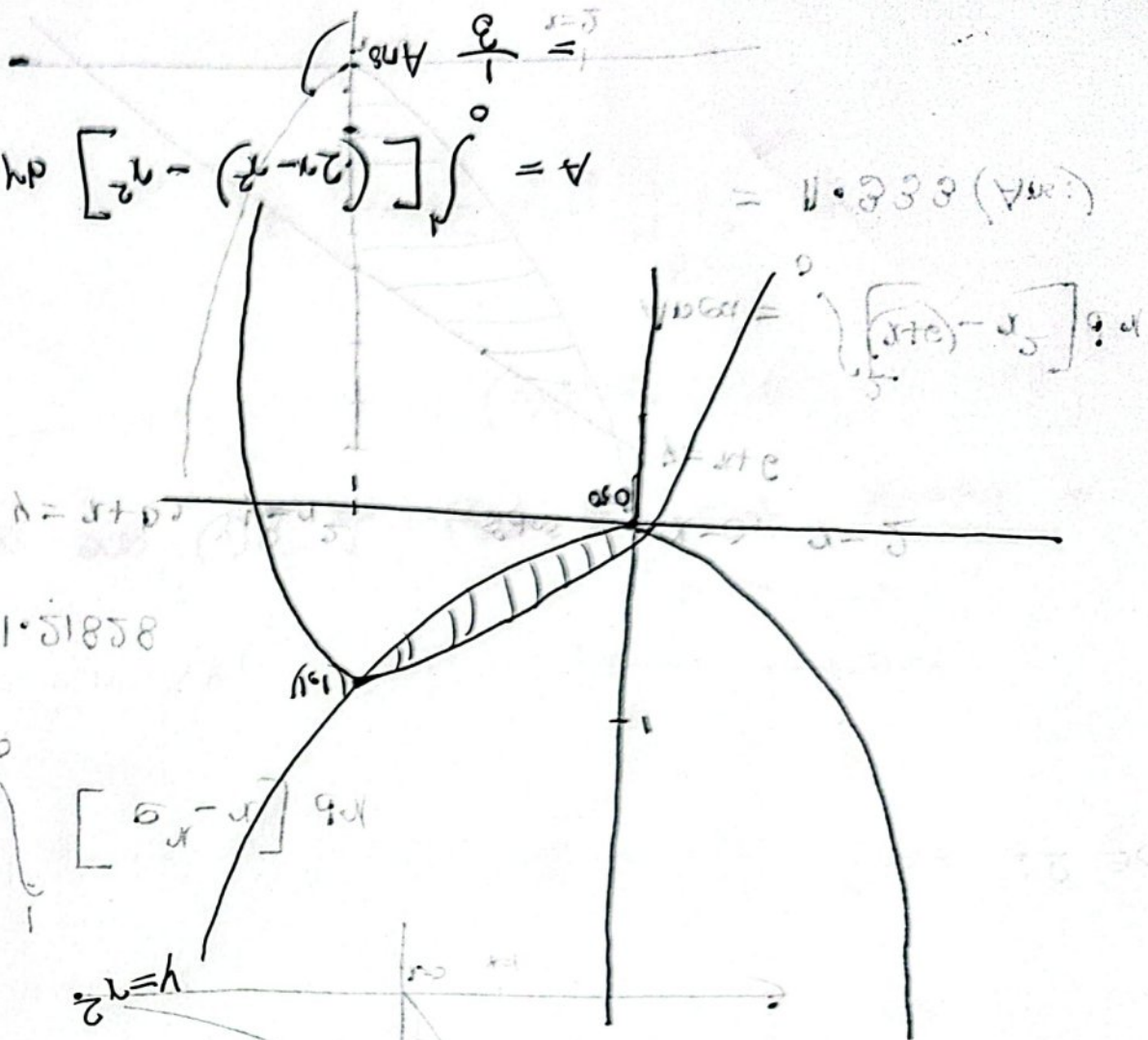
Exp-03

Sketch of the area and find the area of the region

bounded by the parabola $y = x^2$ and the line $y = 2x - x^2$

Soln: $y = x^2$ and $y = 2x - x^2$ are parabolas opening upwards and downwards respectively

$$\begin{aligned}
 &= -(x^2 - 2 \cdot x \cdot 1 + 1^2 - 1) \\
 &= -(x-1)^2 + 1 \\
 &= 1 - (x-1)^2 \\
 &= 1 - x^2 + 2x - 1 \\
 &= -x^2 + 2x
 \end{aligned}$$



$$A = \int_0^2 [(2x - x^2) - x^2] dx$$

$$= \frac{1}{3} \text{ Ans.}$$

$$= 1.5858$$

$$= A$$

$$A = \int_{-2}^4 \left[(y+1) - \left(\frac{y^2-6}{2} \right) \right] dy$$

$$= 18$$

Lec-08

If the limit exists, the improper integral is said to be ~~converge~~ converge otherwise diverge

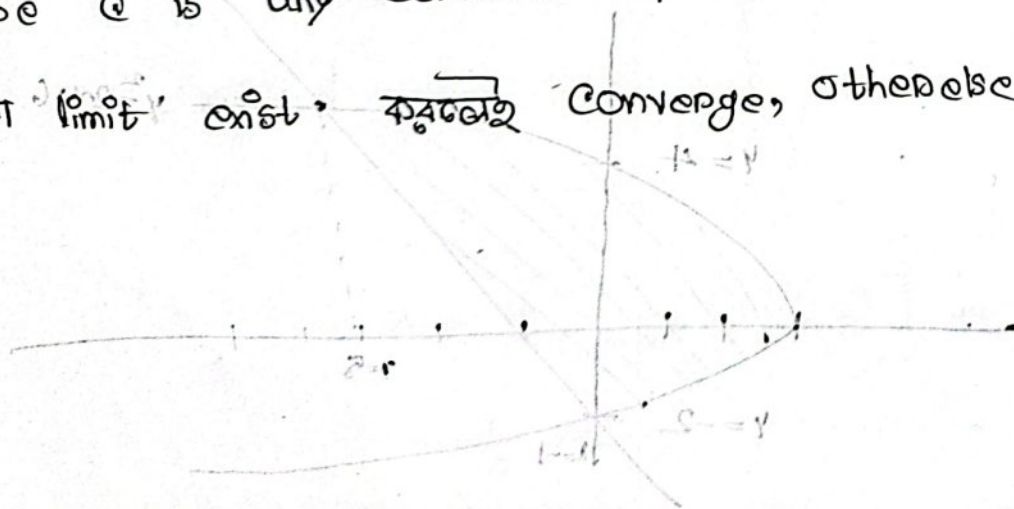
$$① \int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$$

$$② \int_a^{\infty} f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx$$

$$③ \int_{-\infty}^{\infty} f(x) dx = \lim_{a \rightarrow -\infty} \int_a^c f(x) dx + \lim_{b \rightarrow \infty} \int_c^b f(x) dx$$

where c is any convenient point

→ If limit exist, ~~converge~~ Converge, otherwise diverge



Exp-01

$$\int_1^{\infty} \frac{1}{x} dx$$

$$\Rightarrow \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x} dx$$

$$\Rightarrow \lim_{b \rightarrow \infty} [\ln x]_1^b$$

$$\Rightarrow \lim_{b \rightarrow \infty} [\ln b - \ln 1]$$

$$\Rightarrow \lim_{b \rightarrow \infty} \ln b$$

$$\Rightarrow \infty \text{ [Diverge]}$$

Exp-03

$$\int_a^{\infty} (1-x)e^{-x} dx \quad \text{--- ①}$$

$$\Rightarrow \int (1-x)e^{-x} dx$$

$$\Rightarrow (1-x) \int e^{-x} dx - \left\{ \frac{d}{dx} (1-x) \int e^{-x} dx \right\}$$

$$\Rightarrow \left[(1-x) \frac{e^{-x}}{-1} \right] - \int \left\{ (-1) \frac{e^{-x}}{-1} \right\} dx$$

Exp-02

$$\int_2^{\infty} \frac{1}{(x-1)^2} dx$$

$$\Rightarrow \lim_{b \rightarrow \infty} \int_2^b \frac{1}{(x-1)^2} dx$$

$$\Rightarrow \lim_{b \rightarrow \infty} \left[(x-1)^{-2+1} \right]_2^b$$

$$\Rightarrow \lim_{b \rightarrow \infty} \left[\frac{(x-1)^{-2+1}}{(-2+1) \cdot 1} \right]_2^b$$

$$\Rightarrow \lim_{b \rightarrow \infty} \left[\frac{-1}{(x-1)} \right]_2^b$$

$$\Rightarrow \lim_{b \rightarrow \infty} \left[\frac{-1}{b-1} - \left(\frac{-1}{2-1} \right) \right]$$

$$\Rightarrow 0+1 = 1 \text{ [converge]}$$

$$\Rightarrow \left[-(1-x) \cdot e^{-x} \right] - \left[\frac{e^{-x}}{-1} \right]$$

$$\Rightarrow \left[-(1-x)e^{-x} \right] + e^{-x}$$

$$\Rightarrow -e^{-x} + xe^{-x} + e^{-x}$$

No al,

$$\int_0^{\infty} (1-x) e^{-x} dx = \lim_{b \rightarrow \infty} \int_0^b (1-x) e^{-x} dx$$

$$\Rightarrow \lim_{b \rightarrow \infty} \left[\frac{x e^{-x}}{(1-x)} \right]_0^b \quad \text{nil} \Rightarrow \infty \leftarrow b$$

$$\Rightarrow \lim_{b \rightarrow \infty} (b e^{-b} - 0) \quad \text{nil} \Rightarrow \infty \leftarrow b$$

$$\Rightarrow \lim_{b \rightarrow \infty} (b e^{-b}) \quad \text{nil} \Rightarrow \infty \leftarrow b \quad [\text{indeterminate}]$$

$$\Rightarrow \lim_{b \rightarrow \infty} \frac{b}{e^b} \quad [L'Hospital Rule]$$

$$\Rightarrow \lim_{b \rightarrow \infty} \frac{1}{e^b} \quad \text{nil} \Rightarrow \infty \leftarrow b$$

$$\Rightarrow \lim_{b \rightarrow \infty} \left(\frac{1}{e^b} - \frac{1}{e^b} \right) \quad \text{nil} \Rightarrow \infty \leftarrow b$$

$$\neq \frac{1}{e^{\infty}} \quad 0+1=1 \quad \text{converge}$$

$$\Rightarrow \left[\frac{1}{e^b} - \frac{1}{e^b} \right] - \left[\frac{1}{e^b} - \frac{1}{e^b} \right] \quad \text{converge}$$

$$\Rightarrow \sigma^{-g} [\text{converge}]$$

$$r^{-g} + r^{-g} + r^{-g} + \dots$$

$$\left\{ r^{-g} (r-1) \frac{b}{r-1} \right\} - r b r^{-g} (r-1)$$

$$r b \left\{ \frac{r^{-g}}{r-1} (r-1) \right\} - \left[\frac{r^{-g}}{r-1} (r-1) \right]$$

Ex-01

$$\int_0^{\infty} \frac{1}{x} dx$$

$$\int_0^{\infty} \frac{1}{x} dx \quad \text{nil} \Rightarrow \infty \leftarrow b$$

$$\int_0^{\infty} [x dx] \quad \text{nil} \Rightarrow \infty \leftarrow b$$

$$[x dx - x dx] \quad \text{nil} \Rightarrow \infty \leftarrow b$$

$$x dx \quad \text{nil} \Rightarrow \infty \leftarrow b$$

$$[Diverge] \quad \infty \Rightarrow \infty \leftarrow b$$

Ex-02

$$\int_0^{\infty} (1-x) e^{-x} dx$$

$$\int_0^{\infty} r^{-g} (r-1) dx$$

Exp-04

Test the convergence. $\int_1^4 \frac{dx}{(x-2)^{2/3}}$

Soln:

The integral is improper because the integrand approaches $+\infty$ at $x=2$.

$$\text{So, } \int_1^4 \frac{dx}{(x-2)^{2/3}} = \int_1^2 \frac{dx}{(x-2)^{2/3}} + \int_2^4 \frac{dx}{(x-2)^{2/3}}$$

$$\Rightarrow \lim_{b \rightarrow 2} \int_1^b \frac{dx}{(x-2)^{2/3}} + \lim_{a \rightarrow 2} \int_a^4 \frac{dx}{(x-2)^{2/3}} \quad \text{--- (i)}$$

$$\begin{aligned} \therefore \int \frac{dx}{(x-2)^{2/3}} &= \frac{(x-2)^{-2/3+1}}{(-\frac{2}{3}+1)} = 3(x-2)^{1/3} \\ \lim_{b \rightarrow 2} [3(x-2)^{1/3}]_1^b + \lim_{a \rightarrow 2} [3(x-2)^{1/3}]_a^4 &= [3(b-2)^{1/3} - 3(1-2)^{1/3}] + [3(4-2)^{1/3} - 3(a-2)^{1/3}] \\ &\Rightarrow 6.7 \text{ [converge]} \end{aligned}$$

Exp-05

$$\int_{-\infty}^{\infty} \frac{e^{-x}}{1+e^{-2x}} dx$$

$$\Rightarrow \int_{-\infty}^0 \frac{e^{-x}}{1+e^{-2x}} dx + \int_0^{\infty} \frac{e^{-x}}{1+e^{-2x}} dx$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \left[\frac{x^{2/3}}{2/3} \right]_1 + \lim_{x \rightarrow 0^+} \left[\frac{x^{2/3}}{2/3} \right]_K$$

$$\Rightarrow \frac{3}{2} \lim_{x \rightarrow 0^+} \left[x^{2/3} - (-1)^{2/3} \right] + \frac{3}{2} \lim_{x \rightarrow 0^+} \left[x^{2/3} - K^{2/3} \right]$$

$$\Rightarrow \frac{3}{2} (0 - 1) + \frac{3}{2} (4 - 0) = \frac{3}{2} (-1 + 4) = \frac{3}{2} (3) = \frac{9}{2}$$

$\Rightarrow$ Converge

Lec Note: RH Riemann

Beta-Gamma

The Gamma Function (Improper Integral Application)

The Gamma Function is defined by the integral,

$$\Gamma(n) = \int_0^{\infty} e^{-x} \cdot x^{n-1} dx$$

Properties:

$$\Rightarrow \Gamma(n+1) = n \Gamma(n) = n!$$

$$\Rightarrow \Gamma(1) = 1$$

$$\Rightarrow \Gamma(1/2) = \sqrt{\pi}$$

The Beta Function is defined by the integral

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad [m, n > 0]$$

Properties :

$$\Rightarrow B(m, n) = B(n, m)$$

$$\Rightarrow \int_0^{\pi/2} \sin^p \theta \cos^q \theta \, d\theta = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right) \frac{\pi}{2} + (1-0) \frac{\pi}{2} = \frac{\pi}{2}$$

$$\Rightarrow \Gamma m \cdot \Gamma - m = \frac{\pi}{\sin(m\pi)}$$

Relⁿ Betⁿ Beta and Gamma Function's val

$$B(m, n) = \frac{\Gamma m \Gamma n}{\Gamma m+n}$$

Exp-1

$$\Gamma n = \int_0^{\infty} e^{-x} x^{n-1} dx$$

$$= \int_0^{\infty} e^{-x} x^4 dx$$

$$= \int_0^{\infty} e^{-x} x^{5-1} dx$$

$$= \frac{1}{5}$$

$$\left[0 < \frac{1}{5} < \sqrt{4+1} \right] = 4!$$

Exp-2

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Q1 $\int_0^1 x^5 (1-x)^6 dx$

$$\Rightarrow \int_0^1 x^{6-1} (1-x)^{7-1} dx$$

$$\Rightarrow B(6, 7)$$

$$\Rightarrow \frac{6! 7!}{12!}$$

$$\Rightarrow \frac{5! 6!}{12!}$$

$$\Rightarrow \frac{5! 6!}{12!}$$

* $n! = n \cdot (n-1)! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 1$

$$\sqrt{\frac{3}{2}} = \sqrt{\frac{3}{2} + 1} = \frac{3}{2} \sqrt{\frac{1}{2}} = \frac{3}{2} \sqrt{\frac{1}{2} + 1} = \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}}$$

$$\Rightarrow \frac{3}{2} \times \frac{1}{2} \sqrt{\frac{1}{2}} = \frac{3}{4} \sqrt{\frac{1}{2}}$$

$$\therefore \sqrt{\frac{5}{2}} = \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}} = \frac{3}{4} \sqrt{\frac{1}{2}}$$

$$\sqrt{\frac{7}{2}} = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}} = \frac{15}{8} \sqrt{\frac{1}{2}}$$

Exp-3

$$\int_0^{\pi/2} \sin^3 \theta \cos^4 \theta d\theta$$

$$\Rightarrow \frac{1}{2} \beta\left(\frac{3+1}{2}, \frac{4+1}{2}\right)$$

$$\Rightarrow \frac{1}{2} \beta\left(2, \frac{5}{2}\right)$$

$$\Rightarrow \frac{1}{2} \frac{\Gamma(2) \cdot \Gamma(\frac{5}{2})}{\Gamma(\frac{9}{2})}$$

$$\Rightarrow \frac{1}{2} \frac{\Gamma(1+1) \Gamma(\frac{3}{2}+1)}{\Gamma(\frac{7}{2}+1)}$$

$$\Rightarrow \frac{1}{2} \frac{1! \cdot \frac{3}{2} \Gamma(\frac{3}{2})}{\frac{7}{2} \Gamma(\frac{7}{2})}$$

$$\Rightarrow \frac{1}{2} \cdot \frac{\frac{3}{2} \Gamma(\frac{1}{2}+1)}{\frac{7}{2} \Gamma(\frac{5}{2}+1)}$$

$$\Rightarrow \frac{1}{2} \frac{\frac{3}{2} \cdot \frac{1}{2} \Gamma(\frac{1}{2})}{\frac{7}{2} \cdot \frac{5}{2} \Gamma(\frac{3}{2})}$$

$$\Rightarrow \frac{1}{2} \frac{\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{\frac{7}{2} \cdot \frac{5}{2} \sqrt{\frac{3}{2}}}$$

$$\Rightarrow \frac{1}{2} \frac{\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \sqrt{\frac{2}{2}}}$$

$$\Rightarrow \frac{1}{2} \frac{\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \sqrt{\frac{1}{2}+1}}$$

$$\Rightarrow \frac{1}{2} \frac{\frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}}}$$

$$\Rightarrow \frac{1}{2} \frac{\frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}}{\frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}}$$

$$= \frac{2}{35} \text{ (Ans.)}$$

$$|a| = \sqrt{a^2} = \sqrt{1+a^2} \quad (*)$$

$$\frac{a}{c} = \frac{\sqrt{a^2}}{c} = \frac{\sqrt{1+\frac{a^2}{c^2}}}{\frac{c}{c}} = \frac{\sqrt{1+\frac{a^2}{c^2}}}{1}$$

$$\sqrt{1+\frac{a^2}{c^2}} \Rightarrow \sqrt{1+\frac{a^2}{c^2}} \Rightarrow \sqrt{1+\frac{a^2}{c^2}}$$

$$\sqrt{1+\frac{a^2}{c^2}} = \frac{1}{c} \sqrt{1+\frac{a^2}{c^2}} = \frac{\sqrt{1+\frac{a^2}{c^2}}}{c}$$

$$\sqrt{1+\frac{a^2}{c^2}} = \frac{1}{c} \sqrt{1+\frac{a^2}{c^2}} = \frac{\sqrt{1+\frac{a^2}{c^2}}}{c}$$

Exp-04

$$\int_0^{\pi/2} \cos^6 \theta \cdot d\theta$$

$$\Rightarrow \int_0^{\pi/2} \sin^0 \theta \cos^6 \theta d\theta = \frac{\sin^1 \theta}{1} = \sin \theta$$

$\Rightarrow$ $\frac{\pi/2}{2} \beta \left(\frac{9+6}{2} \right)$

$$\Rightarrow \frac{1}{2} \cdot \beta\left(\frac{0+1}{2}, \frac{6+1}{2}\right)$$

$$\Rightarrow \frac{1}{2} \beta \left(\frac{1}{2}, \frac{7}{2} \right)$$

$$\Rightarrow \frac{1}{2} \frac{\sqrt{\frac{1}{2}} \sqrt{\frac{7}{2}}}{\sqrt{4}}$$

$$\Rightarrow \frac{1}{2} \frac{\sqrt{\pi} \left(\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} \right)}{\sqrt{3+1}}$$

$$\Rightarrow \frac{1}{2} \frac{15\sqrt{\pi}}{3!}$$

$$\Rightarrow \frac{\frac{1}{2} \cdot \frac{15}{8} \sqrt{\pi}}{6}$$

$$\Rightarrow \frac{1}{2} \times \frac{15}{8} \sqrt{\pi} \times \frac{1}{6}$$

$$\Rightarrow \frac{5\pi}{32} \text{ (Ans.)}$$

$\frac{1}{2} \left(\frac{1}{2} \right)^n \cdot \left(\frac{1}{2} \right)^n$

$$\left. \begin{array}{l} \text{200 (22) } \\ \text{(28x2) } \end{array} \right\} \quad \text{200}$$

$$56 \frac{1}{6} (\Sigma)^{100} (\Sigma)^{100} \left(\frac{1}{100} \right) \left(\frac{1}{100} \right)$$

$$\lim_{x \rightarrow \infty} \left(\frac{x}{x^2 + 1} \right)^{\frac{1}{x}} = \frac{1}{e}$$

$$\Rightarrow \frac{1}{2} \left(\sin 2\alpha + \cos 2\alpha \right)$$

$$\sin^2 \alpha \leq \cos^2 \alpha \leq \sin^2 \beta \quad \Rightarrow \quad \frac{\pi}{6} \leq \alpha \leq \frac{\pi}{3}$$

$$\Rightarrow \frac{7}{5} \times \frac{1}{5} \cdot \left(\frac{14}{5}, \frac{14}{5} \right)$$

$$\left(\frac{1}{2}, \frac{1}{2}\right) \text{ 9 } \frac{1}{2} \times \frac{1}{2} =$$

$$\frac{\frac{1}{C} \times \frac{H}{E}}{\frac{1}{C} \times \frac{H}{E}} \in$$

$$\frac{100}{100} \leq \frac{\sqrt{\frac{1}{2} \times \frac{3}{2} \times \frac{3}{2} \times \sqrt{\frac{1}{2}}}}{14} \leq \frac{14}{2}$$

সেই ক্ষেত্রে $z = 3\theta$

$$\frac{dz}{d\theta} = 3$$

$$dz = 3d\theta$$

$$d\theta = \frac{1}{3} dz$$

θ	0	$\pi/6$
z	0	$\pi/2$

$$\left(\frac{1+\cos \theta}{2} \right) \theta \quad \frac{1}{2}$$

$$\left(\frac{\pi}{2} \cos \frac{1}{2} \right) \theta \quad \frac{1}{2}$$

$$\frac{\sqrt{\pi} \sqrt{\frac{1}{2}}}{\sqrt{2}} \quad \frac{1}{2}$$

$$\frac{\left(\sqrt{\pi} \sqrt{\frac{1}{2}} \cdot \frac{\pi}{2} \cdot \frac{1}{2} \right) \sqrt{\pi}}{\sqrt{2}} \quad \frac{1}{2}$$

$$\frac{\sqrt{\pi} \sqrt{\frac{1}{2}}}{\sqrt{2}} \quad \frac{1}{2}$$

$$\frac{\sqrt{\pi} \sqrt{\frac{1}{2}}}{2} \quad \frac{1}{2}$$

$$\frac{1}{2} \times \sqrt{\pi} \sqrt{\frac{1}{2}} \times \frac{1}{2}$$

$$(\text{ans}) \quad \frac{\pi \sqrt{2}}{16}$$

$$5* \int_0^{\pi/6} \sin^2(6\theta) \cdot \cos^4(3\theta) d\theta$$

$$\Rightarrow \int_0^{\pi/6} \sin^2(2 \times 3\theta) \cos^4(3\theta) d\theta$$

$$\Rightarrow \int_0^{\pi/2} \sin^2(2z) \cos^4(z) \frac{1}{3} dz$$

$$\Rightarrow \frac{1}{3} \int_0^{\pi/2} \{\sin^2(2z)\}^2 \cos^4 z dz$$

$$\Rightarrow \frac{1}{3} \int_0^{\pi/2} (2 \sin z \cos z)^2 \cos^4 z dz$$

$$\Rightarrow \frac{4}{3} \int_0^{\pi/2} \sin^2 z \cos^6 z dz$$

$$\Rightarrow \frac{4}{3} \times \frac{1}{2} \beta\left(\frac{2+1}{2}, \frac{6+1}{2}\right)$$

$$\Rightarrow \frac{4}{3} \times \frac{1}{2} \beta\left(\frac{3}{2}, \frac{7}{2}\right)$$

$$\Rightarrow \frac{4}{3} \times \frac{1}{2} \frac{\Gamma(3/2) \Gamma(7/2)}{\Gamma(4+1)}$$

$$\Rightarrow \frac{4}{6} \frac{\frac{1}{2}\sqrt{\pi} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2}\sqrt{\pi}}{4!} \Rightarrow \frac{5\pi}{192}$$

Q1 $\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \pi$

$\Rightarrow \int_0^{\pi/2} \sqrt{\frac{\sin \theta}{\cos \theta}} d\theta$

∞	0	x
$-\infty$	0	Σ

$\Rightarrow \int_0^{\pi/2} \sin^{1/2} \theta \cdot \cos^{-1/2} \theta d\theta$

$\Rightarrow \frac{1}{2} B\left(\frac{\frac{1}{2}+1}{2}, \frac{-\frac{1}{2}+1}{2}\right)$

$\Rightarrow \frac{1}{2} B\left(\frac{3}{4}, \frac{1}{4}\right)$

$\Rightarrow \frac{1}{2} \frac{\sqrt{\frac{3}{4}} \sqrt{\frac{1}{4}}}{\sqrt{\frac{4}{4}}}$

$\Rightarrow \frac{1}{2} \sqrt{\frac{1}{4}} \left(\sqrt{1-\frac{1}{4}}\right)$

~~$\Rightarrow \frac{1}{2} \sqrt{\frac{1}{4}}$~~

$\Rightarrow \frac{1}{2} \times \frac{\pi}{\sin(\frac{1}{4}\pi)} \Rightarrow \frac{1}{2} \times \frac{\pi}{\sin(\frac{\pi}{4})}$

$\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \pi$

$\sqrt{x} = \sum_{n=0}^{\infty} \binom{1/2}{n} x^n$
 $\Rightarrow \frac{1}{\sqrt{x}} = \sum_{n=0}^{\infty} \binom{-1/2}{n} x^n$
 $\Rightarrow \frac{1}{\sqrt{x}} = \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{4^n n!} x^n$
 $\Rightarrow \frac{1}{\sqrt{x}} = \sum_{n=0}^{\infty} \frac{(-1)^n (2n-1)!!}{4^n n!} x^n$

$\int_0^{\pi/2} \sqrt{\tan \theta} d\theta = \int_0^{\pi/2} \tan^{1/2} \theta d\theta$
 $\int_0^{\pi/2} \tan^{1/2} \theta d\theta = \int_0^{\pi/2} \sin^{-1/2} \theta \cos^{-1/2} \theta d\theta$
 $\int_0^{\pi/2} \sin^{-1/2} \theta \cos^{-1/2} \theta d\theta = \frac{1}{2} B\left(\frac{1}{2}, \frac{1}{2}\right)$
 $\frac{1}{2} B\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2} \frac{\Gamma(\frac{1}{2}) \Gamma(\frac{1}{2})}{\Gamma(1)}$
 $\frac{1}{2} \frac{\Gamma(\frac{1}{2}) \Gamma(\frac{1}{2})}{\Gamma(1)} = \frac{1}{2} \frac{\sqrt{\pi} \sqrt{\pi}}{1} = \frac{\pi}{2}$

$\frac{\pi}{2} \times \frac{1}{\sin(\frac{\pi}{4})} = \frac{\pi}{2} \times \frac{1}{\frac{1}{\sqrt{2}}} = \frac{\pi}{2} \times \sqrt{2} = \frac{\pi\sqrt{2}}{2}$

Q1 $\int_0^{\infty} e^{-\sqrt{x}} x^{1/4} dx$ Let

$\Gamma = \int_0^{\infty} e^{-x} x^{n-1} dx$

$\Rightarrow$ Let, $z = \sqrt{x}$

$\Rightarrow dz = \frac{1}{2\sqrt{x}} dx$

$\Rightarrow 2dz = \frac{1}{\sqrt{x}} dx$

$\Rightarrow 2\sqrt{x} dz = dx$

$\Rightarrow 2z dz = dx$

x	0	∞
z	0	∞

$\therefore \int_0^{\infty} e^{-z} (z^2)^{1/4} 2z dz$

$\Rightarrow 2 \int_0^{\infty} e^{-z} z^{1/2} z dz$

$\Rightarrow 2 \int_0^{\infty} e^{-z} z^{3/2} dz$

$\Rightarrow 2 \int_0^{\infty} e^{-z} z^{\frac{5}{2}-1} dz$

$\Rightarrow 2 \sqrt{\frac{5}{2}} \Rightarrow 2 \times \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi}$
 $= \frac{3}{2} \sqrt{\pi}$

$\left(\frac{1}{\mu} - \frac{e}{\mu} \right) \frac{1}{2}$

$\frac{\frac{1}{\mu} - \frac{e}{\mu}}{\frac{1}{\mu}}$

$\left(\frac{1}{\mu} - 1 \right) \frac{1}{4}$

$\frac{\pi}{\left(\frac{\pi}{\mu} \right)^{n/2}} \times \frac{1}{2} = \frac{\pi}{\left(\pi \frac{1}{\mu} \right)^{n/2}} \times \frac{1}{2}$

Q1 $\int_0^{\infty} e^{-x^2} dx$

Let,

$$z = x^2$$

$$dz = 2x dx$$

$$\Rightarrow \frac{dz}{2\sqrt{z}} = dx$$

x	0	∞
$(\frac{z}{2} - i)$	0	∞

$$\Rightarrow \int_0^{\infty} e^{-x^2} dx = \int_0^{\infty} e^{-z} \frac{1}{2\sqrt{z}} dz$$

$$\Rightarrow \frac{1}{2} \int_0^{\infty} e^{-z} z^{-1/2} dz \Rightarrow \frac{1}{2} \int_0^{\infty} e^{-z} z^{\frac{1}{2}-1} dz$$

$$\Rightarrow \frac{1}{2} \times \Gamma_{1/2} = \frac{1}{2} \sqrt{\pi} = \frac{\sqrt{\pi}}{2}$$

Q1 $\int_0^{\infty} x^6 e^{-2x} dx$

Let,

$$z = 2x$$

$$dz = 2 dx$$

$$dx = \frac{dz}{2}$$

$$\Rightarrow \int_0^{\infty} \left(\frac{z}{2}\right)^6 e^{-z} \frac{dz}{2}$$

$$\Rightarrow \frac{1}{64 \times 2} \int_0^{\infty} e^{-z} z^6 dz$$

$$\Rightarrow \frac{1}{128} \Gamma_7 = \frac{1}{128} \Gamma_{6+1} = \frac{1}{128} \times$$

$6!$	$\frac{45}{8}$	7
1	8	7

Q1 $\int_0^2 \frac{x^2}{\sqrt{2x-x^2}} dx$

$= \int_0^2 \frac{x^2}{\sqrt{2x(1-\frac{x}{2})}} dx$

$= \frac{1}{\sqrt{2}} \int_0^2 x^{2-\frac{1}{2}} \frac{1}{\sqrt{1-\frac{x}{2}}} dx$

$\Rightarrow \frac{1}{\sqrt{2}} \int_0^2 x^{3/2} \left(1-\frac{x}{2}\right)^{-\frac{1}{2}} dx$

Let,

$z = \frac{x}{2}$

~~$dz = \frac{1}{2} dx$~~

~~$\Rightarrow 2 dz = dx$~~

$\Rightarrow 2z = x$

$\Rightarrow \frac{d}{dx} 2z = \frac{d}{dx} x$

$\Rightarrow 2 dz = dx$

x	0	2
z	0	1

$\frac{1}{\sqrt{2}} \int_0^1 (2z)^{3/2} (1-z)^{-\frac{1}{2}} dz$

$\Rightarrow 4 \int_0^1 z^{3/2} (1-z)^{-\frac{1}{2}} dz$

$\Rightarrow 4 \int_0^1 z^{5/2-1} (1-z)^{\frac{1}{2}-1} dz$

$4 \times B\left(\frac{5}{2}, \frac{1}{2}\right)$

Exp-3

Integration formula

$$\int_0^a y^z \sqrt{a^4 - y^4} dy \quad [a \text{ is a constant number}]$$

$$\Rightarrow \int_0^a y^z \sqrt{a^4 \left(1 - \frac{y^4}{a^4}\right)} dy$$

$$\Rightarrow \int_0^a y^z a^2 \sqrt{1 - \frac{y^4}{a^4}} dy$$

$$\Rightarrow a^2 \int_0^a y^z \left(1 - \frac{y^4}{a^4}\right)^{\frac{1}{2}} dy$$

$$\Rightarrow a^2 \int_0^a y^z (1-z)^{\frac{1}{2}} \frac{a^4}{4y^3} dz$$

$$\Rightarrow a^2 \int_0^a y^3 \cdot y^4 (1-z)^{\frac{1}{2}} \frac{a^4}{4y^3} dz$$

$$\Rightarrow \frac{a^6}{4y^3} \int_0^a y^4 (1-z)^{\frac{1}{2}} dz$$

$$\Rightarrow \frac{a^6}{4} \int_0^a z a^4 (1-z)^{\frac{1}{2}} dz$$

$$\Rightarrow \frac{a^6}{4} \int_0^a z^{2-1} (1-z)^{1-\frac{1}{2}} dz$$

$$\text{Let, } z = \frac{y^4}{a^4} \Rightarrow I = \int_0^1 \dots$$

$$\Rightarrow \frac{dz}{dy} = \frac{1}{a^4} \cdot 4y^3$$

$$\Rightarrow a^4 dz = 4y^3 dy$$

$$\Rightarrow dz = \frac{4y^3}{a^4} dy$$

$$\Rightarrow dy = \frac{a^4 dz}{4y^3}$$

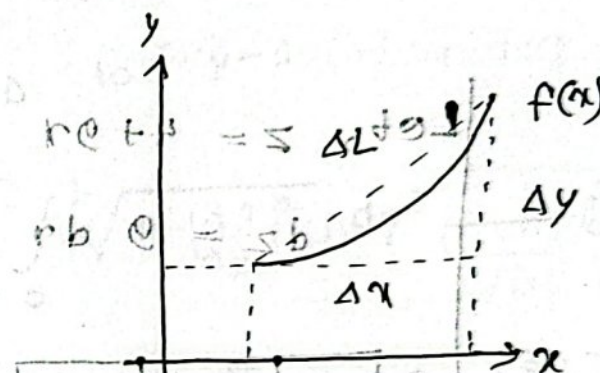
y		
z		

Lec-10
18/07/20

$$[v]_B = r \cdot \text{norm}$$

Topic $\Rightarrow$ Arc Length

Arc Length is the actual distance measured along a curve between two points.



We know that,

Δ	1	r
Δ	Δ	Δ

$$(\Delta L)^2 = (\Delta x)^2 + (\Delta y)^2$$

$$\Rightarrow (\Delta L)^2 = (\Delta x)^2 \left[1 + \left(\frac{\Delta y}{\Delta x} \right)^2 \right]$$

$$\Rightarrow \left(\frac{\Delta L}{\Delta x} \right)^2 = 1 + \left(\frac{\Delta y}{\Delta x} \right)^2$$

$$\Rightarrow \frac{\Delta L}{\Delta x} = \sqrt{1 + \left(\frac{\Delta y}{\Delta x} \right)^2}$$

$$\Rightarrow \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta L}{\Delta x} \right) = \lim_{\Delta x \rightarrow 0} \sqrt{1 + \left(\frac{\Delta y}{\Delta x} \right)^2}$$

$$\Rightarrow \frac{dL}{dx} = \sqrt{1 + \left(\frac{dy}{dx} \right)^2}$$

$$\Rightarrow dL = \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx$$

$$\therefore L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx} \right)^2} dx \quad [y = f(x)]$$

Similarly, [When $x = g(y)$]

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

Topic $\Rightarrow$ Area Length

01-001
20/03/20

Exp (1) Find the arc length of $y = x^{3/2}$ from (0,0) to $(2, 2\sqrt{2})$

Soln:

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx} (x^{3/2}) \\ &= \frac{3}{2} x^{3/2-1} \\ &= \frac{3}{2} x^{1/2} \end{aligned}$$

$$L = \int_0^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= \int_0^2 \sqrt{1 + \left(\frac{3}{2} x^{1/2}\right)^2} dx$$

$$= \int_0^2 \sqrt{1 + \frac{9}{4} x} dx$$

$$= \frac{1}{2} \int_0^2 \sqrt{4 + 9x} dx$$

Let $z = 4 + 9x$

$dz = 9 dx$

x	1	2
z	13	22

$\Rightarrow \Delta z = 9 \Delta x$

$$L = \frac{1}{2} \int_{13}^{22} \sqrt{z} \frac{dz}{9}$$

$$= \frac{1}{2} \times \frac{1}{9} \int_{13}^{22} \sqrt{z} dz$$

$$= \frac{1}{2} \times \frac{1}{9} \int_{13}^{22} z^{1/2} dz$$

$$= \frac{1}{18} \left[\frac{z^{3/2}}{3/2} \right]_{13}^{22}$$

$[x] = \lfloor x \rfloor$

$$\Rightarrow \frac{1}{18} \int_3^{22} \frac{z^{3/2}}{3/2}$$

$$= 2.08 \text{ (Ans.)}$$

Exp-02

Find the arc length of $x = \frac{1}{3}(y^2+2)^{3/2}$

from $y=0$ to $y=1$

$$\text{Soln: } L = \int_0^1 \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$\Rightarrow \frac{dx}{dy} = \frac{d}{dy} \left[\frac{1}{3} (y^2+2)^{3/2} \right]$$

$$\Rightarrow \frac{1}{3} \cdot \frac{3}{2} (y^2+2)^{1/2} \cdot 2y$$

$$= y(y^2+2)^{1/2}$$

$$\Rightarrow L = \int_0^1 \sqrt{1 + \{y(y^2+2)^{1/2}\}^2} dy$$

$$= \int_0^1 \sqrt{1 + y^4 + 2y^2} dy$$

$$= \int_0^1 \sqrt{(y^2+1)^2} dy$$

$$\Rightarrow L = \int_0^1 (y^2 + 1) dy$$

$$= \left[\frac{y^3}{3} + y \right]_0^1$$

$$= \left(\frac{1}{3} + 1 \right) = \frac{4}{3} \text{ (Ans.)}$$

Topic: Arc Length for parametric curve:

$$L = \int_{t=a}^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Exp (1)

Find the arc

length of a parametric

curve

$$x = \cos 2t, y = \sin 2t, 0 \leq t \leq \pi/2$$

Soln:

$$L = \int_0^{\pi/2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$= \int_0^{\pi/2} \sqrt{(-2\sin 2t)^2 + (2\cos 2t)^2} dt$$

$$= \int_0^{\pi/2} \sqrt{4(\sin^2 2t + \cos^2 2t)} dt$$

$$\Rightarrow 2 \int_0^{\pi/2} dt \Rightarrow 2 [t]_0^{\pi/2} = 2 \times \frac{\pi}{2} = \pi \text{ (Ans.)}$$

Exp-2

Find the arc length of a parametric curve

$$x = \cos t + t \sin t, y = \sin t - t \cos t, 0 \leq t \leq \pi$$

Soln:

$$L = \int_0^{\pi/2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\Rightarrow \int_0^{\pi/2} \sqrt{(-\sin t + \cos t)^2 + (\cos t + \sin t)^2} dt$$

$$\Rightarrow \int_0^{\pi/2} \sqrt{(\cos t - \sin t)^2 + (\cos t + \sin t)^2} dt$$

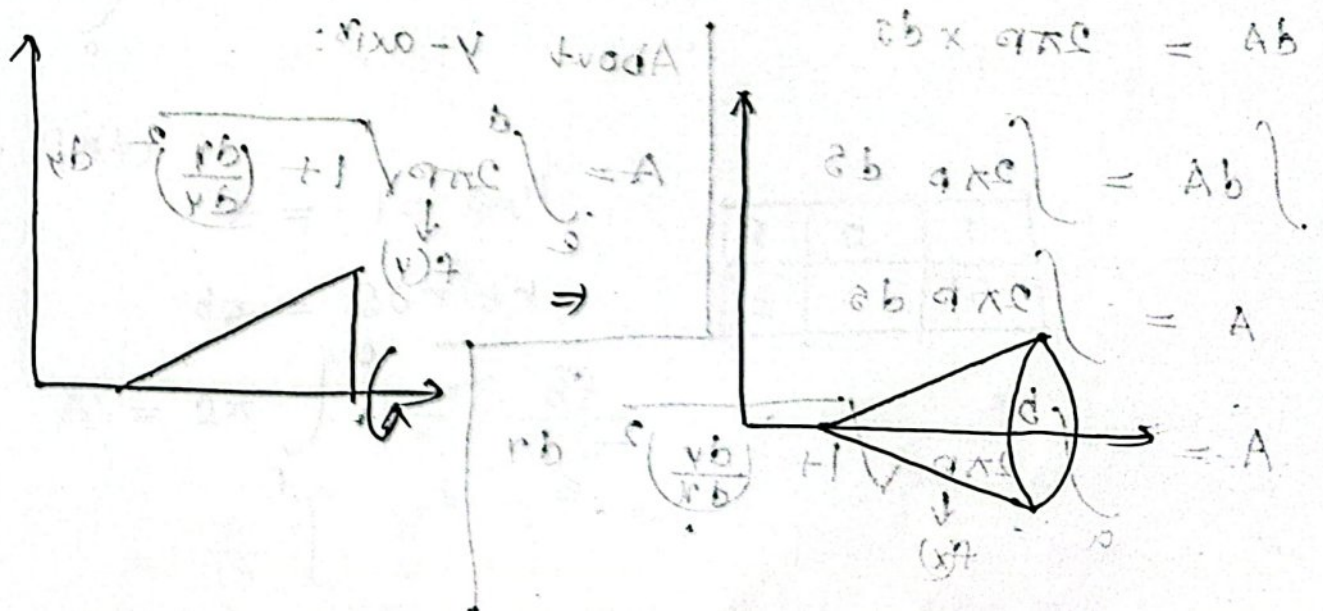
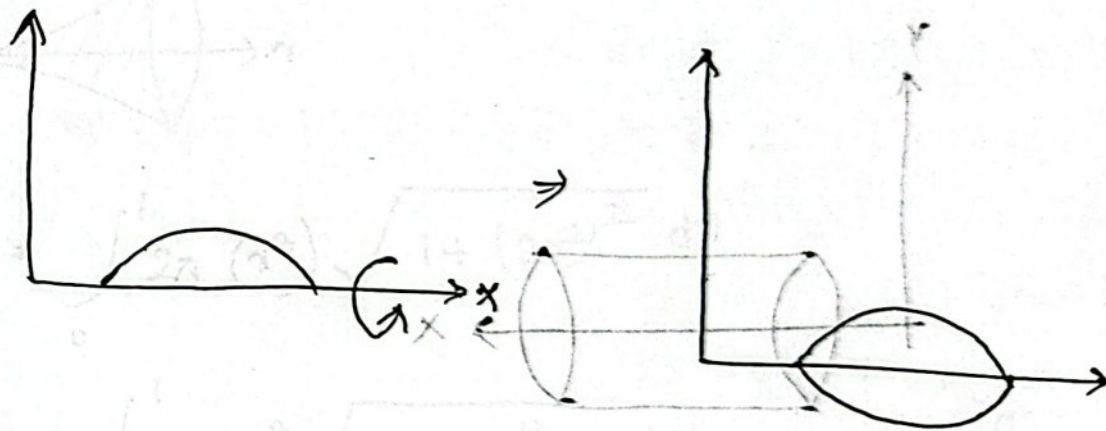
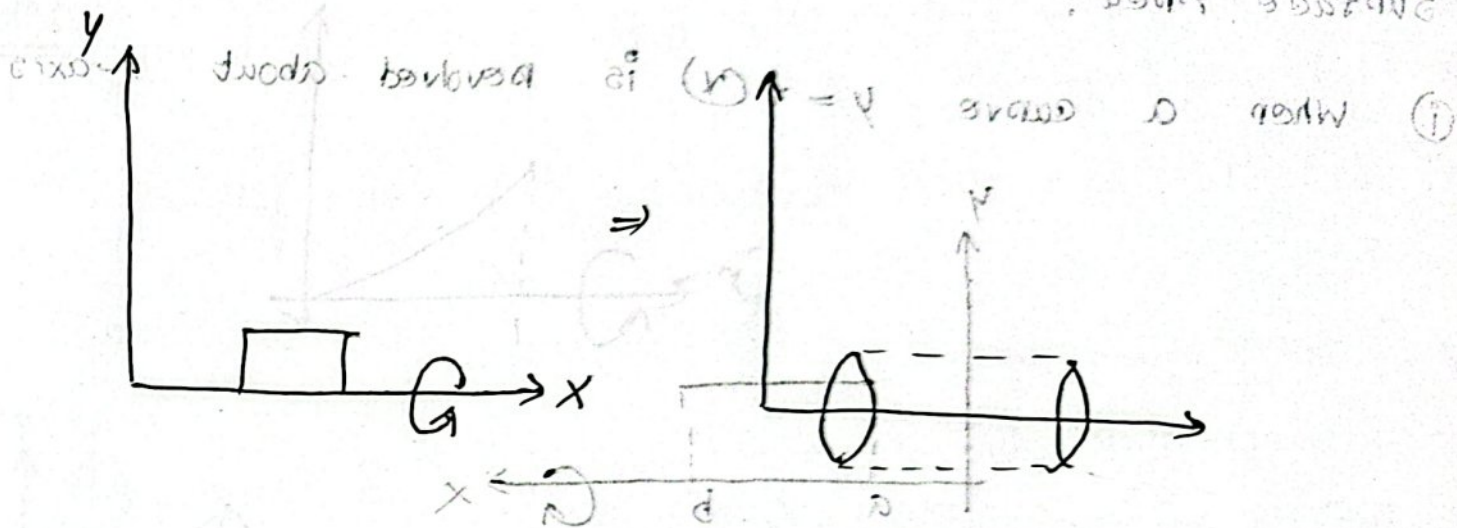
$$\Rightarrow \int_0^{\pi/2} \sqrt{2(\cos^2 t + \sin^2 t)} dt$$

$$\Rightarrow \int_0^{\pi/2} \sqrt{\cos^2 t - 2\cos t \sin t + \sin^2 t + \cos^2 t + 2\cos t \sin t + \sin^2 t} dt$$

$$\Rightarrow \int_0^{\pi/2} \sqrt{2(\cos^2 t + \sin^2 t)} dt$$

$$\Rightarrow \int_0^{\pi/2} \sqrt{2} dt$$

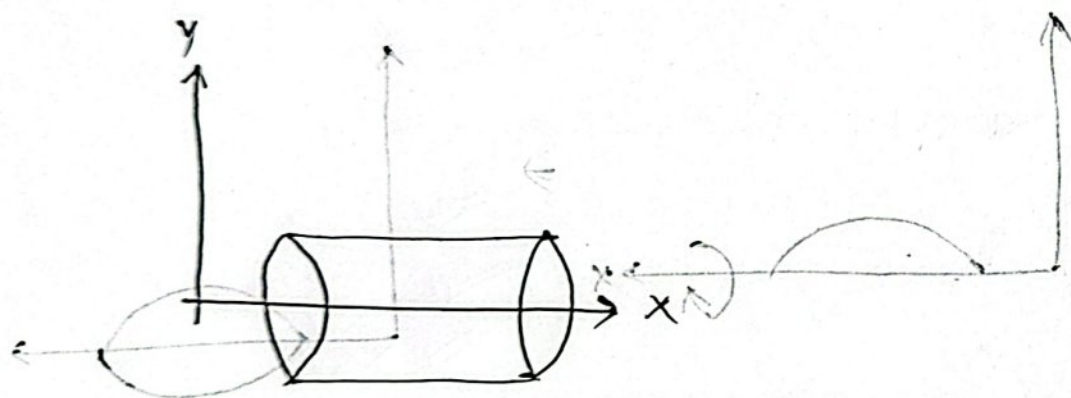
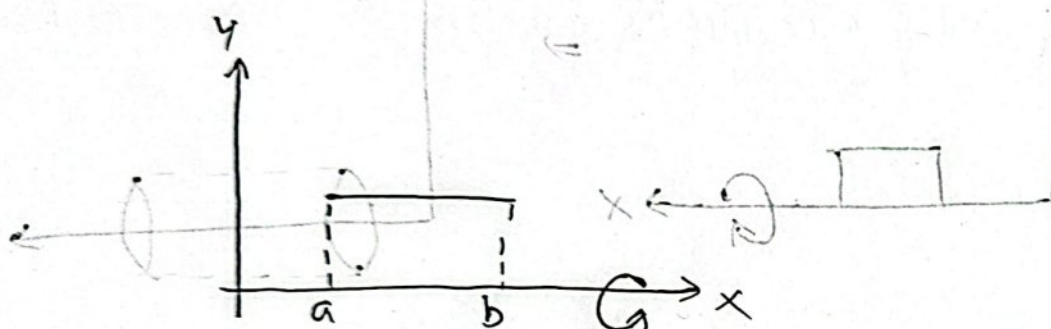
Solid of revolution: is a solid that is generated by revolving a plane region about a line that lies in the same plane.



A surface of revolution is the 3-dimensional surface formed when a plane curve is rotated about a fixed line (axis).

Surface Area:

(i) When a curve $y = f(x)$ is revolved about x -axis



$$dA = 2\pi r \times ds$$

$$\int dA = \int 2\pi r \, ds$$

$$A = \int 2\pi r \, ds$$

$$A = \int_a^b 2\pi f(x) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

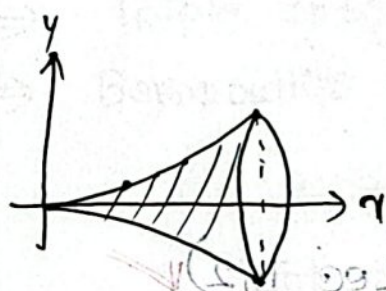
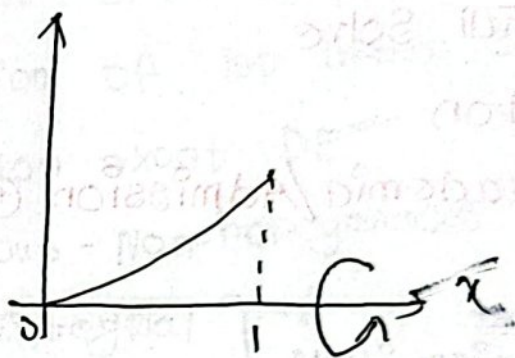
About y -axis:

$$A = \int_a^b 2\pi f(y) \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$



Exp-01 Find the area of the surface that is generated by revolving the portion of the curve $y = x^3$ betⁿ $x=0$ & $x=1$ about the x -axis

Solⁿ:



$$A = \int_0^1 2\pi (x^3) \sqrt{1 + (3x^2)^2} dx$$

$$= \int_0^1 2\pi x^3 \sqrt{1 + 9x^4} dx$$

⇒ Let,

$$z = 1 + 9x^4$$

$$dz = 36x^3 dx$$

$$A = 2\pi \int_0^1 \sqrt{z} \frac{dz}{36}$$

$$= \frac{\pi}{18} \left[\frac{z^{3/2}}{3/2} \right]_1^2$$

2	0	1
z	1	2